

Routledge
Taylor & Francis Group

Sociological Spectrum

Mid-South Sociological Association

ISSN: 0273-2173 (Print) 1521-0707 (Online) Journal homepage: www.tandfonline.com/journals/usls20

Intersection of race, immigration and space: quantifying cultural accessibility for New York City neighborhoods

Qiyao Pan & Juliette Zhu

To cite this article: Qiyao Pan & Juliette Zhu (21 May 2026): Intersection of race, immigration and space: quantifying cultural accessibility for New York City neighborhoods, Sociological Spectrum, DOI: 10.1080/02732173.2026.2666403

To link to this article: https://doi.org/10.1080/02732173.2026.2666403


Published online: 21 May 2026.


Submit your article to this journal


View related articles


View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=usls20

SOCIOLOGICAL SPECTRUM
https://doi.org/10.1080/02732173.2026.2666403

Routledge
Taylor & Francis Group

Check for updates

Intersection of race, immigration and space: quantifying cultural accessibility for New York City neighborhoods

Qiyao Pan and Juliette Zhu

Sociology, The Graduate Center, City University of New York, New York, New York, USA

ABSTRACT

Despite extensive research on spatial access to social services, scholars tend to overlook cultural organizations as a component of neighborhood institutional infrastructure. Access to cultural resources is particularly important for newcomers to navigate and reshape social boundaries. This study explores the spatial dimensions of cultural organizations at the neighborhood level in New York City, which we conceptualize as "cultural accessibility" through two measures: the number of organizations within a 1-mile radius (spatial availability) and the distance to the nearest organization (spatial proximity). Using spatial analysis and regression models, we focus on the intersection of neighborhoods' ethnoracial composition and concentrated immigration. Our findings reveal divergent patterns: higher immigrant concentration reduces access in White neighborhoods, but in Hispanic and Asian neighborhoods, it increases access. Black neighborhoods exhibit lower cultural accessibility, which is largely explained by socioeconomic disadvantage and residential mobility. These patterns persist in access to citywide organizations, while ethnic-oriented organizations enhance access across immigrant neighborhoods of diverse ethnoracial types. Our study furthers the understanding of spatial inequalities in cultural access and advocates for greater attention to it role in shaping opportunities for immigrant communities.

Introduction

How does space intersect with organizations to shape residents' access to social capital in their neighborhoods? Coleman (1988) indicated that social capital constitutes resources available through one's social ties depending on the size of network connections. Scholars often study social capital in space with a focus on the location and concentration of social networks in a neighborhood (Small and Adler 2019). In particular, researchers have examined the relationship between space and networks with attention to different compositions of networks in a neighborhood, such as healthcare services (Roubenoff, Slootjes, and Bloemraad 2023; Tadmon and Bearman 2023; Tao et al. 2018; Zhang, Hou, and Liu 2025), childcare centers (Small 2006, 2010), nonprofit organizations (Edwards and McCarthy 2004; Hwang and Lee 2023) and parks (Cheng et al., 2022; Elliott, Korver–Glenn, and Bolger 2019; Kawlra and Sakamoto 2023; Winkler et al. 2024). Scholars focusing on social services emphasize the positive role of organizations in network building and resource mobilization (Edwards and McCarthy 2004; Hwang and Lee 2023; Small 2010). Like social services, cultural organizations foster social connections and civic engagement (DiMaggio and Fernandes-Kelly 2010). More importantly, as Bourdieu (1986) points out, they play a foundational role in reproducing culture capital, which in turn shapes identity

CONTACT Qiyao Pan qpan@gradcenter.cuny.edu Candidate in Sociology, The Graduate Center, City University of New York, New York, New York, USA.
© 2026 Taylor & Francis Group, LLC

and social status. Access to cultural resources is particularly important for newcomers (DiMaggio and Fernandes-Kelly 2010; Kasinitz et al. 2009). In immigrant gateways like New York City, cultural organizations, ranging from world-class museums to local arts centers, serve as critical arenas where immigrants can navigate and reshape social boundaries (Alba 2024; DiMaggio and Fernandes-Kelly 2010; Kasinitz 2023; Kasinitz and Martiniello 2019). Cultural organizations also play a vital role in immigrant's everyday life, making a space where the boundaries between insiders and outsiders blur (Alba 2024; DiMaggio and Fernandes-Kelly 2010; Kasinitz and Martiniello 2019).

Cultural organizations are also an indispensable part of network ties, and like other services, spatial proximity is essential to facilitate interactions among residents for them to benefit from the network connections (Foster et al. 2015). In urban contexts, cultural organizations can drive neighborhood change by attracting the "creative class," leading to both revitalization and potential gentrification (Bridge 2006; Florida 2003). While social services, such as healthcare and childcare centers, are receiving most scholarly attention, we know very little about how cultural organizations influence residents' access to social capital at the neighborhood level. Particularly, the spatial dimensions of cultural organizations in neighborhoods remain elusive. Only until recently have some researchers begun to focus on the spatial dimensions of social capital and the rise of creative class in urban neighborhoods (Foster et al. 2015; Graif 2018), which motivates this study to examine cultural access from a spatial perspective. Although the difficulty in determining which level to be measured complicates the place-based conceptualizations of social capital in general (Foster et al. 2015; Sampson and Graif 2009), a place-based approach to social networks should incorporate spatial thinking and proximity measurement (Logan 2012).

In this study, we investigate the spatial dimensions of cultural organizations at the neighborhood level in New York City. Our overall research question is: how does the access to cultural organizations differ by a neighborhood's ethnoracial composition and immigrant concentration? Quantifying cultural accessibility is an important step in understanding how people access specific sources and networks in space as a form of structural social capital (Foster et al. 2015). We adopt Galster and Sharkey (2017)'s *spatial opportunity structure* framework to investigate how spatial access to cultural organizations varies across New York City's diverse neighborhoods. We operationalize neighborhoods as census tracts, the smallest and consistently defined geographical units commonly used in urban research to approximate local residential environments (Logan et al. 2011). Census tracts are designed to be homogenous in population and socioeconomic characteristics (U.S. Census Bureau 2024), and their boundaries follow natural boundaries of streets, rivers or other limits to reflect local neighborhoods (Graif 2018).

We conceptualize cultural accessibility as the spatial opportunity structure through which residents may encounter and engage with cultural institutions—that is, the degree to which cultural organizations are available and reachable from a given neighborhood. We operationalize this concept along two dimensions: 1) *spatial availability*, measured as the number of cultural organizations within a 1-mile radius; and 2) *spatial proximity*, measured as the average distance to the nearest cultural organization. Together, these two measures capture complementary aspects of cultural accessibility: availability reflects the density of cultural resources in a neighborhood's surroundings, while proximity captures how close residents are to the nearest cultural organization. Drawing on data from the New York City Department of Cultural Affairs (DCLA) directory, we follow DCLA's definitions and consider cultural organizations as New York City-based nonprofit entities that provide arts and cultural services to the public, including visual, performing and literary arts, public-oriented science and humanities institutions such as museums, zoos and botanical gardens. In our analysis, we also address the heterogeneity of cultural organizations by considering two specific types of organizations: 1) *ethnic-oriented* organizations that explicitly serve racial, ethnic or immigrant communities; and 2) *citywide* organizations that serve broader mainstream audiences across the city.

Our agenda is threefold: First, we quantify the spatial accessibility to cultural organizations by analyzing how the number of organizations within surrounding areas and the distance to

nearest cultural organizations differ by neighborhoods. Second, we examine how cultural accessibility varies by ethnoracial composition and concentrated immigration at the neighborhood level. Third, we examine how a neighborhood's socioeconomic status and residential mobility may mediate the outcome of cultural accessibility. We apply spatial analysis and multiple regression models to address these questions.

Our analysis reveals divergent patterns: higher immigrant concentration reduces access in White neighborhoods, but in Hispanic and Asian neighborhoods, it increases access. In contrast, Black neighborhoods exhibit lower cultural accessibility, a pattern that is largely explained by socioeconomic status and residential mobility. To some extent, our findings echo the idea that Asians and Hispanic neighborhoods follow distinctive trajectories from Black neighborhoods shaped by large-scale immigration and lower level of segregation (Alba 2024; Logan and Stults 2021). Furthermore, ethnic-oriented organizations primarily enhance cultural access by proximity among immigrant neighborhoods, whereas citywide organizations remain spatially stratified and mirror the overall patterns of cultural accessibility.

This study employs the lens of intersectionality to investigate how ethnoracial composition, immigration concentration, and cultural accessibility are interrelated at the neighborhood level and together shape spatial inequalities in access to cultural resources. Our analysis contributes to the current understanding of immigrant and racial communities in a superdiverse city (Vertovec 2007) by highlighting the blurred boundaries of cultural accessibility across neighborhood types. We also highlight cultural accessibility as a dimension of the assimilation and integration experience for Hispanic and Asian immigrants, and advocate for greater scholarly attention to the crucial role of cultural access in shaping opportunities for immigrant communities.

Literature review

Cultural organizations and neighborhood context

The extensive literature has established the significant role of organizations in shaping how individuals access social capital, form network ties, and organize collective actions in their neighborhoods (Bridge 2006; Foster et al. 2015; Jorgensen and Stedman 2011; Lizardo and Jilbert 2023; Ruef and Kwon 2016; Rutten, Westlund, and Boekema 2010; Sampson 2012; Sampson and Graif 2009; Small and Adler 2019; Zilberstein 2019). Social capital scholars have examined various types of organizations, including financial institutions (Small et al. 2021), healthcare facilities (Roubenoff et al. 2023; Tadmon and Bearman 2023; Tao et al. 2018; Zhang et al. 2025), childcare centers (Small 2006, 2010), nonprofit organizations (Edwards and McCarthy 2004; Hwang and Lee 2023), and parks (Cheng et al. 2022; Elliott et al. 2019; Kawlra and Sakamoto 2023; Winkler et al. 2024). While social capital theorists emphasize the neighborhood context of these organizations in fostering network ties and mobilizing resources, they often overlook the cultural dimensions as an essential component of social capital.

In contrast, scholars often situate cultural organizations within the context of urbanism and meaning making when examining the relationship between culture and place (Borer 2006; Clark 2003; Lloyd 2004, 2010; Zukin 1982, 1996; Zukin et al. 2009). Cultural organizations are often conceptualized as engines of economic development and neighborhood change in metropolis like New York City and Chicago (Clark 2003; Currid-Halkett 2007; Florida 2003). Specifically, the growth of cultural organizations can attract "creative class" as both residents and consumers, and the increase of middle-class residents contributes to the gentrification of the neighborhoods (Florida 2003; Zukin et al. 2009). While scholars rooted in the tradition of political economy have criticized the commodification of authenticity associated with cultural offerings, cultural organizations may also benefit neighborhoods in other ways. Recent studies have shown that the existence of cultural organizations is associated with reduced neighborhood disadvantage (Foster et al. 2015), and greater neighborhood diversity (Graif 2018).

Moreover, neighborhood effects scholars highlight the role of cultural resources as important dimensions of resident-based social capital and collective efficacy (Foster et al. 2015; Sampson 2012; Sampson and Graif 2009). In his landmark work on Chicago neighborhoods, Sampson (2012) argued that neighborhoods influence a remarkably wide range of social phenomena and events through different types of community-based organizations. To measure organizations' neighborhood effects, Sampson created an indicator: the density of organization-based resources, which was a multi-item index of the mean number of survey-reported local organizations and program resources in a neighborhood such as community newspapers, neighborhood watches, alcohol/drug treatment programs, family planning centers, afterschool recreational programs, counseling and mental health clinics, etc. On the other hand, despite Sampson acknowledging the importance of cultural activities in shaping collective actions in neighborhoods, the density of cultural organizations and resources was not included in the multi-item index to measure neighborhood effects and inequalities.

Recently, neo-Chicago School scholars have also examined the connections between cultural amenities and neighborhoods, emphasizing their contemporary symbolic meaning and their role as cultural forces in urban transformation (Clark 2003; Grazian 2008; Ocejo 2017; Silver and Clark 2016). In particular, Silver and Clark (2016) developed a theory of "scenes" to quantify the diverse cultural meanings conveyed through the lifestyles associated with specific amenities. Specifically, they argued that cultural organizations can be mapped and analyzed as components of the scenes (Silver and Clark 2016). While they advocated using amenities as indicators to measure urban scenes, their emphasis on the cultural meanings paid less attention to how access to these amenities might vary across neighborhoods along ethnoracial lines.

Building on the work by Sampson (2012) and Silver and Clark (2016), our study addresses this gap by examining the spatial dimensions of cultural organizations at the neighborhood level, with a focus on ethnoracial inequalities in access and proximity. We extend the analysis to the interaction between ethnoracial composition and immigrant concentration of a neighborhood, to offer a more nuanced understanding of the different trajectories of cultural accessibility. We also adopt a place-based approach to examine the spatial inequalities and the implications of available spatial opportunities for residents and their households.

Racialized spatial accessibility to organizations and services

Why does spatial accessibility matter for organizational resources at the neighborhood level? Galster and Sharkey (2017) argued that the spatial organization of inequality directly and indirectly shapes individual attributes, labor market outcomes and access to opportunities across the life course. To a large extent, spatial inequality is organized through deliberate efforts to maintain and reinforce existing inequalities, as evidenced by the persistent racial segregation (Dreier, Mollenkopf, and Swanstrom 2014; Galster and Sharkey 2017; Massey and Denton 1988). On the other hand, the spatial opportunity structures also provide pathways for social mobility and well-being (Galster and Sharkey 2017). Spatial accessibility to cultural organizations facilitates access to social capital for residents, especially for those who lack the resources and mobility to access services outside of their neighborhoods (Curley 2010).

In recent years, researchers began to study the spatial dimensions of social capitals, networks, and other resources through a place-based approach (Downey 2003; Foster et al. 2015; Galster and Sharkey 2017; Graif 2018; Jorgensen and Stedman 2011; Logan 2012; Rutten et al. 2010; Small 2006; Small et al. 2021). Specifically, sociologists focus on proximity to services and goods at the neighborhood level. Lacking access to resources such as high-quality food, education, healthcare, and financial services creates disadvantages for neighborhoods to strive. Small and his colleagues (2021) focused on neighborhoods considered as "banking deserts" and found that physical proximity to a bank and alternative financial institutions affects banking practices. Their research demonstrated a remarkable primacy of race over class in the distribution of financial services across neighborhoods, that alternative financial institutions are closer than the bank

even in low-poverty Black neighborhoods than high-poverty White ones (Small et al. 2021). The racialized prevalent access to alternative financial institutions is an example of how neighborhood service affects racial inequality.

Other studies also revealed the racial disparities in spatial accessibility to different types of neighborhood resources, in addition to financial services (Elliott et al. 2019; McKenzie 2013; Rhubart et al. 2022; Sampson 2012). Elliott et al. (2019) used the longitudinal data from 1947 to 2015 to study access to parks during the urbanization process in Houston. Their findings not only confirmed that socially privileged residents enjoyed greater access to parks but also suggested that minorities have improved their access to parks through moving into historically White neighborhoods, rather than creating new parks. Using data from the 2000 Census and the 5–year 2005–2009 ACS, McKenzie (2013) investigated the unequal transit access across neighborhoods in Portland, Oregon, and found that Latino concentrated neighborhoods had the poorest relative access to transit, and the access level even declined in Latino and Black concentrated neighborhoods during the study period. In their recent study, Rhubart and colleagues (2022) analyzed the sociospatial disparities in third places, finding that poverty is associated with fewer third places, and tracts with the smallest share of Blacks and Hispanics had more third places.

These racialized disparities in neighborhood infrastructure did not emerge in a vacuum. For Black neighborhoods in particular, decades of racial segregation through restrictive covenants, redlining and discriminatory lending practices systematically deprived these communities of public and private investment (Massey and Denton 1988; Rothstein 2018). This concentrated disadvantage in Black neighborhoods is persistent across generations, weakening the organizational capacity of neighborhoods and leaving them with fewer civic and nonprofit organizations (Small 2006; Wilson 1987). In New York City, the urban renewal programs further disrupted community organizations in Black neighborhoods through large-scale demolition and displacement of residents (Fullilove 2016).

Although these studies highlight the racialized access to neighborhood services and amenities, few focus on the cultural dimensions of neighborhood amenities. Building on previous work on spatial proximity to neighborhood services, we adopt the *spatial opportunity structure* framework (Galster and Sharkey 2017) to address the racial and ethnic disparities in the spatial accessibility to cultural organizations. We extend our research with policy implications regarding the potential role of cultural organizations in fostering social capital opportunities.

Immigrant neighborhoods in cultural economy

Since the 1980s, mass waves of immigration from Asia, Latin America and the Caribbean have transformed the urban landscape with their cultural practices. In a superdiverse city like New York City, cultural expressions ranging from music to art to sports, have also become increasingly multiethnic and multiracial (Kasinitz and Martiniello 2019). Contrary to early twentieth-century scholars' belief, that cultural incorporation was solely about Americanization, today's cultural landscapes reflect the strong cultural influences of immigrant cultures (Alba 2024; Foner 2023; Kasinitz and Martiniello 2019). In particular, immigrants have played a significant role in transforming popular culture and arts during the early twentieth-first century (Alba 2024; Kasinitz and Martiniello 2019). From K-pop to Broadway, newcomers and children of immigrants are revitalizing the "American" traditions within the cultural economy (Kasinitz et al. 2009).

Cultural practices are a vital part of immigrant communities, with recent studies indicating that immigrants have become more actively engaged in cultural activities and organizations (DiMaggio and Fernandes-Kelly 2010). Stern et al. (2010) found that the cultural participation rate, as well as the number of nonprofit cultural providers, commercial cultural firms, resident artists, were, in fact, higher in immigrant neighborhoods than in the rest of the metropolitan area in Philadelphia. As they explained, this was mainly because the immigrant block groups

with the highest concentration of cultural participation are those located in or near the city center. Other studies also showed that multiple immigrant groups especially Latin American, African American and Asian immigrants, have incorporated nonprofit, commercial and informal cultural forms into their lives (DiMaggio and Fernandes-Kelly 2010; Kasinitz and Martiniello 2019; Stern, Seifert, and Vitiello 2010). Meanwhile, second-generation immigrants also seek artistic expression as a mean of creativity and self-employment (DiMaggio and Fernandes-Kelly 2010). Children of immigrants often use "expressive entrepreneurship" to overcome market constraints and maintain their cultural identity (DiMaggio and Fernandes-Kelly 2010; Fernández-Kelly 2015).

Despite the vibrant cultural expressions of first- and second-generation immigrants, researchers have rarely investigated disparities in cultural access across immigrant and ethnoracial neighborhoods to explain this phenomenon. Our case study of New York City offers an opportunity to interrogate the intersection between race/ethnicity and immigration, as this superdiverse city is both racially and ethnically diverse with a high concentration of immigrants (Kasinitz 2023; Vertovec 2007).

Data and methodology

Department of Cultural Affairs directory

We obtained the list of all cultural organizations in the Department of Cultural Affairs (DCLA) directory from the NYC Open Data website. The DCLA directory is the most comprehensive and systematically compiled dataset of cultural organizations in New York City. It is updated by DCLA annually and was last updated on June 25, 2021, at the time of extraction. Following DCLA's definitions, we consider cultural organizations as New York City-based nonprofit entities that provide arts and cultural services to the public, including visual, performing and literary arts, as well as public-oriented science and humanities institutions such as museums, zoos and botanical gardens. We focus on nonprofit organizations for two reasons. First, nonprofit cultural organizations are a core component of neighborhood institutional infrastructure. They typically serve place-based constituencies with free or low-cost programming and are more likely to reflect cultural needs of local communities, including ethnoracial minorities and immigrants (Small 2006). For-profit venues and informal cultural spaces operate under different market logics and serve different audiences, which are conceptually distinct from the place-based cultural infrastructure we examine here. Second, the DCLA directory does not include for-profit entities. We further discuss the implications of this scope in the limitations section. Although geographic information is available in the dataset, 19% of the organizations have missing geometries. To solve this issue, we geocoded those organizations with identifiable addresses and dropped those with only post office box information or no identifiable address. The final dataset contains 2030 cultural organizations in New York City by the year 2021, as shown in Figure 1.

Organization type: ethnic-oriented and citywide

To address the heterogeneity of cultural organizations, we also classify two additional categories based on an organization's mission, scope and target clientele: 1) *ethnic-oriented* organizations that explicitly serves racial, ethnic or immigrant communities (n = 137); and 2) *citywide* organizations that serve broader mainstream audiences across the city (n = 754). To implement this classification, we leveraged the GPT-4o model to retrieve publicly available mission statements via Google search on and developed keyword dictionaries for category matching. For *ethnic-oriented* dictionary, we include keywords of ethnic and racial groups, immigrant communities, diaspora and heritage terms; for *citywide* dictionary, we incorporate keywords related to major institutions, prestigious venues, high art programming, libraries and public services. Organizations that did not match either category were retained in the overall set of cultural organizations; because our proximity-based accessibility framework already conceptualizes the local, community-serving

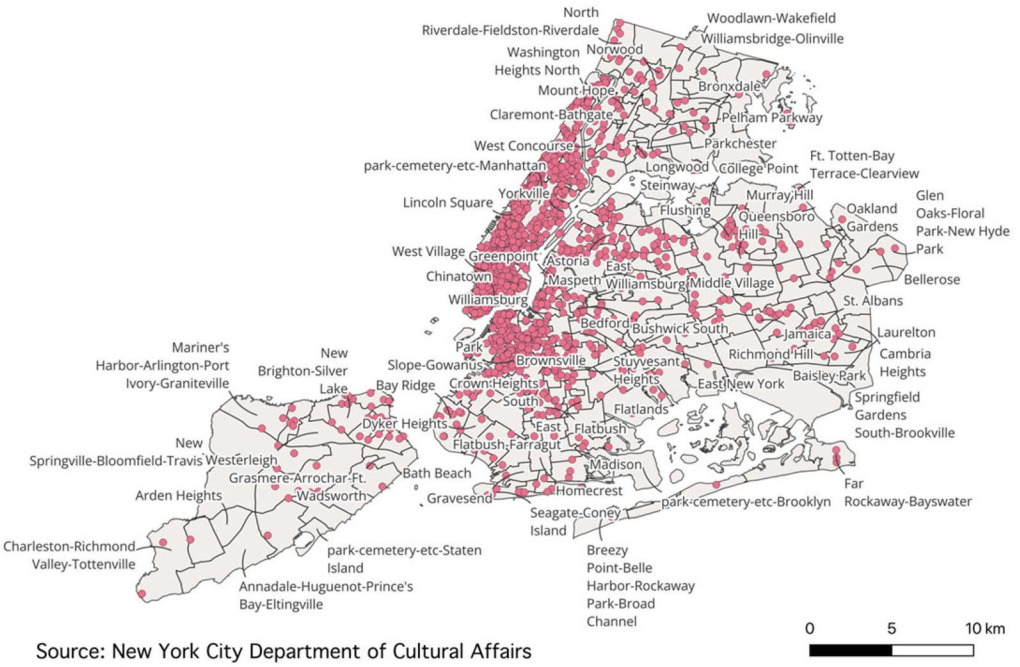


Figure 1. Locations of cultural organizations in New York City.

function of these organizations, we do not model a separate analytic category for unspecified community-based organizations. For further details on the classification methodology, see Appendix 1.

As a robustness check, we manually coded 100 cases following the same coding scheme. The GPT classifier achieved an accuracy of 92%, which is satisfactory compared to existing studies using large language models for coding. Among the misclassifying cases, the GPT model tends to undercount both ethnic and citywide types. Nevertheless, we consider these conservative estimates of ethnic and citywide organizations to be suitable for the category-specific analysis.

American Community Survey

To examine the relationship between cultural accessibility and neighborhood characteristics, we use the 2017–2021 American Community Survey (ACS) 5-year estimates to classify neighborhood types by ethnoracial composition and capture other neighborhood characteristics. Neighborhoods are measured at the census tract level, which is the smallest unit commonly used in literature to approximate neighborhoods (Logan et al. 2011). Tracts also tend to follow natural boundaries of streets, rivers or other limits (Graif 2018).

Dependent variables

Our dependent variables capture two complementary dimensions of cultural accessibility: spatial availability and spatial proximity. For spatial availability, we count the number of cultural organizations within a 1-mile buffer of each census tract centroid. We adopt this threshold based on the "15-minute city concept" proposed by Bruno and colleagues (2024), which defines the radius within which residents can access essential daily needs; we use the 1-mile buffer as 1 mile approximates the distance covered by a 15-minute walk. For spatial proximity, we calculate the Euclidean distance from each tract centroid to its nearest cultural organization. Euclidean

distance serves as a practical approximation of walking distance, and New York City's urban environment makes it a particularly suitable context for this measure. The city's grid-patterned street network, high walkability and extensive pedestrian infrastructure produce a stable ratio between Euclidean and network-based measures; in other words, Euclidean distance closely approximates actual walking routes at the neighborhood level (Apparicio et al. 2008). Together, these two measures capture the breadth of cultural access surrounding a neighborhood and the immediacy of access to the closest organization. We also compute both measures for ethnic-oriented and citywide organizations. We restrict our neighborhood-level analysis to census tracts with a population of at least 100 residents.

Independent variables

The main independent variable is neighborhood typology based on ethnoracial composition consisting of six types: predominantly White, predominantly Black, predominantly Hispanic, predominantly Asian, White mixed and non-White mixed. Following the methodology used in prior studies (Farrell and Lee 2011; Hall, Crowder, and Spring 2015; Kennedy et al. 2021), we define a neighborhood as predominantly White if non-Hispanic Whites account for over 80 percent of the tract population. For a predominantly minority neighborhood, we consider a specific minority group (e.g., non-Hispanic black) with over 50 percent of the total tract population. For White and non-White mixed neighborhoods, we classify a tract as White mixed if the share of non-Hispanic white groups is between 50 percent and 80 percent in the tract, or non-White mixed if the tract does not fit any of the above categories.

The other independent variable is *concentrated immigration*, an index to reflect the concentration of the immigrant population in a neighborhood. Drawing on Sampson et al. (1997) and Zapatka and Tran (2023), we construct *concentrated immigration* using principal component analysis (PCA) based on two characteristics: 1) the percentage of individuals that do not speak English at home and 2) the percentage of foreign-born individuals, with 84.03% of the variation explained by the construct (Cronbach's alpha = 0.81; see PCA biplot in Appendix 2).

Covariates

We also include three covariates consisting of poverty, education, and mobility factors to control for the socioeconomic status of a neighborhood. Specifically, we measure the share of residents living in poverty, the share of residents with a college degree, and the share of residents who moved from a different county within the same state at the tract level.

Analytic techniques

Our analyses proceed in three steps. First, we apply spatial techniques to measure spatial access to cultural organizations within 1 mile, and the distance to the nearest cultural organization. Specifically, we use point-in-polygon approach and created buffers of 1 mile around each neighborhood centroid to calculate the number of cultural organizations within 1 mile. We measure the distance to the nearest organization in Euclidean distance. As the straight-line distance between two points, Euclidean distance is a standard spatial proximity measure widely used in urban research. It is a practical approximation of walking distance for the grid-patterned urban environment of New York City

Second, we visualized the cultural accessibility based on these two measures, as presented in Figures 2 and 3. Additional maps are available in Appendix 3. For sensitivity analysis, we also constructed a discordance map to compare the patterns of two measures (see Appendix 4). Third, we analyze the association between cultural accessibility and neighborhood characteristics using multiple OLS regression models. In this step, we address the intersection of neighborhoods'

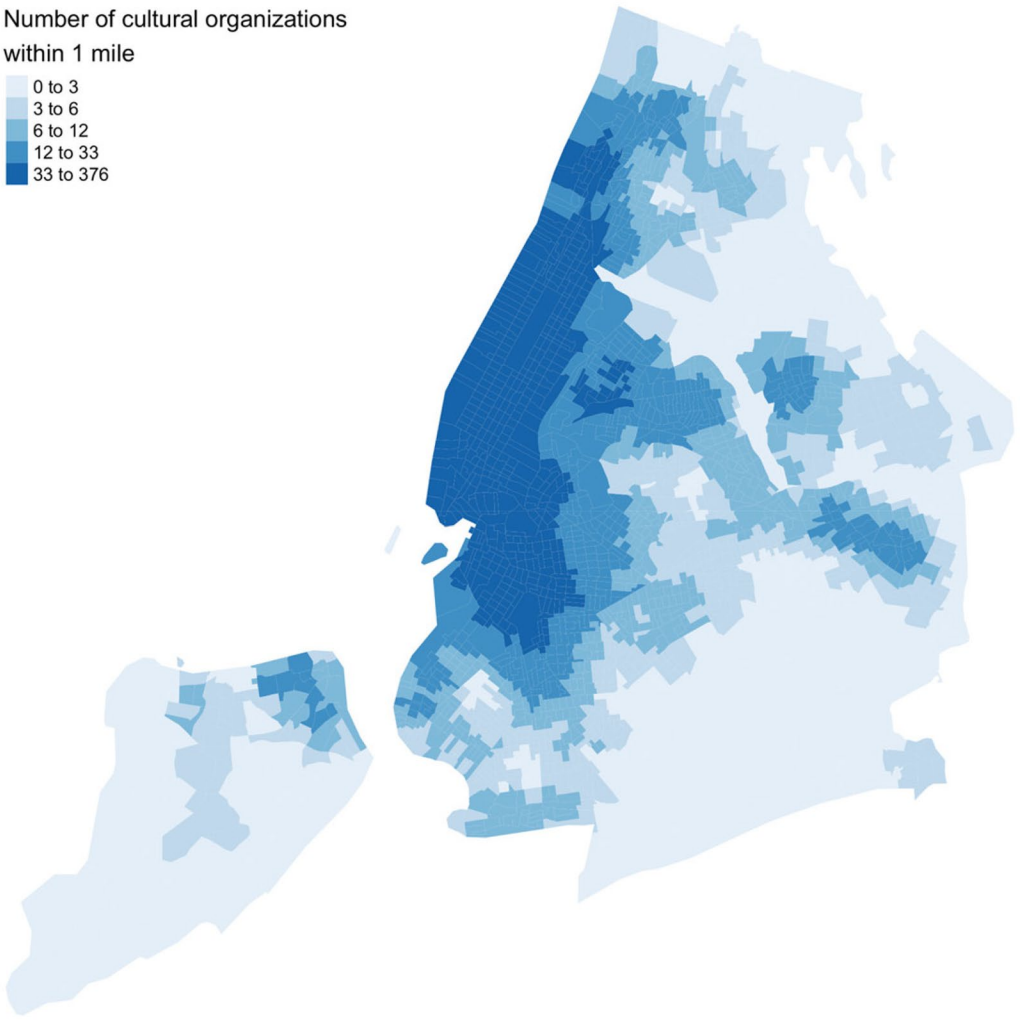


Figure 2. Number of cultural organizations by census tracts in New York City (within 1 mile).

ethnoracial composition and concentrated immigration as the predictors of divergent cultural accessibility. We use OLS as the primary specification because both of our outcome measures are already spatially constructed and therefore incorporate local spillover effect by design. For robustness check, we also estimate spatial lag models using a queen-contiguity weights matrix for main outcomes (see Appendix 5).

Results

Visualizing cultural accessibility

The spatial patterns of cultural organizations and their accessibility in New York City are presented at the tract level in Figure 2 and 3. Figure 2 shows the distribution of cultural organizations within 1 mile by quintile. While it is unsurprising that the number of organizations is concentrated in Manhattan and northern Brooklyn, where most well-known organizations locate in, there is also a high density of cultural organizations in the southern Bronx and a moderate density in eastern and northeast Queens. It also indicates a spillover effect in space, that the density of cultural organizations is also higher in the surrounding neighborhoods

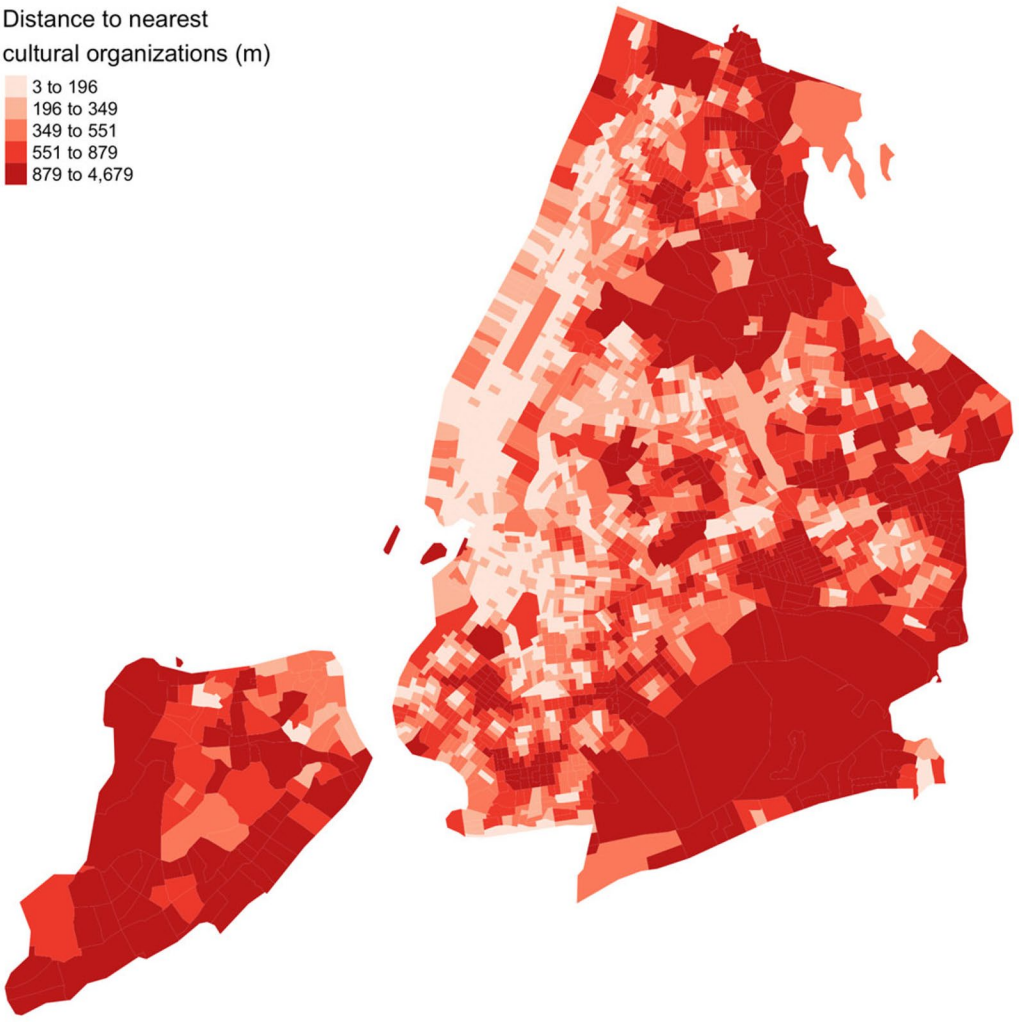


Figure 3. Distance to nearest cultural organizations in New York City.

of the highest-level ones than in other areas. Figure 3 visualizes the cultural accessibility in the nearest distance to cultural organizations, which also shows that most Manhattan neighborhoods and northern Brooklyn enjoy great accessibility. In addition to the airports and stadium arenas in Queens, western Queens and a majority of Staten Islands have low cultural accessibility.

Cultural accessibility by neighborhood typology and immigration

We analyzed how neighborhood typology and immigration are associated with cultural accessibility, and whether socioeconomic and mobility factors account for the relationship. Table 1 summarizes the statistics of all variables. Our analysis includes tracts with at least 100 residents and excluded those missing socioeconomic characteristics in the data, totaling 2105 tracts. On average, there are 31 cultural organizations within 1 mile of a neighborhood, including 2 ethnic-oriented organizations and 11 citywide organizations, and the nearest organization is 566 m away. Among the tracts, 24.5% are White mixed neighborhoods, 21.8% Black, 18.2% Hispanic, 16.5% non-White mixed, 13.6% predominantly White, and 5.4% Asian. Concentrated immigration is standardized, so half of the tracts fall below the mean. For socioeconomic

Table 1. Summary table of cultural accessibility and neighborhood characteristics by census tracts in New York City.

	Overall ($N = 2105$)
Number of organizations within 1 mile (N)	
Mean (*SD*)	31 (62)
Ethnic-oriented	2 (4)
Citywide	11 (26)
Distance to nearest organization	
Mean (*SD*)	566 (474)
Ethnic-oriented	1640 (1340)
Citywide	989 (792)
Neighborhood typology	
Predominantly White	287 (13.6%)
Predominantly Black	458 (21.8%)
Predominantly Hispanic	384 (18.2%)
Predominantly Asian	113 (5.4%)
White Mixed	516 (24.5%)
Non-White Mixed	347 (16.5%)
Concentrated immigration	
Mean (*SD*)	0.00 (1.3)
Median [Min, Max]	−0.2 [−2.5, 5.3]
% Residents in poverty	
Mean (*SD*)	26.4 (16.0)
Median [Min, Max]	22.8 [0, 100]
% Residents with college degree	
Mean (*SD*)	42.5 (19.7)
Median [Min, Max]	38.4 [7.9, 98.0]
County-level residential mobility	
Mean (*SD*)	1.7 (1.9)
Median [Min, Max]	1.09 [0, 18.1]

covariates, the average poverty rate is 26.4%, 42.5% of the residents hold at least a college degree and 1.7% have moved from another county before.

Additionally, we present descriptive statistics on the neighborhood typology in Table 2. As expected, predominantly White and White mixed neighborhoods have a larger concentration of cultural organizations, particularly citywide organizations. However, Hispanic and Asian neighborhoods, alongside White mixed neighborhoods, tend to have shorter distances to nearest organizations. Notably, Asian neighborhoods are closest to ethnic-oriented organizations, which may be explained by their highest concentration of immigrants, followed by non-White mixed neighborhoods and Hispanic neighborhoods, which also have higher levels of immigrant concentration than other neighborhoods. Hispanic neighborhoods, on the other hand, are the second closest to citywide organizations, following White mixed neighborhoods. In addition, Hispanic neighborhoods also have the highest share of residents in poverty, followed by Asian and Black neighborhoods, while White neighborhoods have the highest portion of residents with college degrees. Asian neighborhoods exhibit the lowest rate of county-level mobility. These divergent patterns indicate that cultural accessibility for Hispanic and Asian neighborhoods may be explained more by immigrant concentration than socioeconomic conditions.

To examine how a neighborhood's ethnoracial type and immigrant concentration predict cultural accessibility, we conducted a set of OLS regression models. Table 3 presents how neighborhood types and concentrated immigration predict the amount of access to cultural organizations within 1 mile of a neighborhood. In Model 1, we find that Asian and White mixed neighborhoods have a greater amount of access compared to predominantly White neighborhoods, while Black neighborhoods have a smaller amount of access. Concentrated immigration is a negative predictor for access. In Model 2, after introducing control variables for socioeconomic and mobility factors, and the relationship became more nuanced. When accounting for these factors, all other types of neighborhoods demonstrate greater access compared to predominantly White neighborhoods. Furthermore, poverty and education are found to be significantly positive

Table 2. Descriptive statistics on neighborhood typology.

	Predominantly White	Predominantly Black	Predominantly Hispanic	Predominantly Asian	White Mixed	Non-White Mixed
Organizations within 1 mile	44.4 (80.0)	14.8 (22.4)	15.1 (15.9)	21.9 (53.1)	55.5 (88.4)	26.6 (53.0)
Ethnic-oriented	2.4 (4.7)	1.3 (2.1)	1.3 (1.8)	2.2 (4.3)	3.3 (5.1)	2.2 (3.5)
Citywide	17.6 (33.2)	4.2 (7.2)	4.2 (5.2)	6.8 (16.0)	22.0 (38.8)	9.2 (20.9)
Distance to nearest organization	643.0 (560.0)	629.2 (491.7)	493.8 (383.3)	517.4 (340.7)	451.0 (435.2)	683.6 (506.5)
Ethnic-oriented	2861.7 (3508.2)	2067.5 (1356.1)	1468.7 (811.9)	1220.8 (655.2)	1514.9 (1430.8)	1460.9 (1045.1)
Citywide	1139.6 (1052.2)	1106.2 (717.6)	849.5 (530.8)	945.0 (567.1)	815.7 (791.0)	1136.3 (864.0)
Concentrated immigration	−1.0 (1.1)	−0.6 (0.6)	0.7 (1.1)	2.4 (1.0)	−0.3 (1.1)	0.5 (1.1)
% White	87.7 (5.0)	10.9 (9.0)	28.6 (16.6)	21.6 (11.0)	66.6 (8.4)	32.4 (13.4)
% Black	1.9 (2.0)	74.7 (13.6)	21.2 (13.8)	2.9 (4.0)	5.7 (6.1)	20.9 (16.6)
% Hispanic	10.9 (9.0)	16.6 (12.9)	66.7 (10.1)	15.0 (8.7)	17.9 (10.7)	27.9 (12.2)
% Asian	5.7 (3.9)	3.1 (3.5)	6.7 (8.2)	63.8 (10.2)	17.2 (10.0)	25.2 (15.6)
% Poverty	17.3 (15.4)	27.6 (16.8)	40.2 (13.8)	30.1 (12.5)	18.8 (11.5)	27.1 (12.5)
% College	56.7 (22.0)	33.6 (11.1)	25.8 (10.0)	36.2 (10.6)	58.6 (18.0)	38.9 (13.4)
% Mobility	1.6 (1.9)	1.3 (1.6)	1.6 (1.6)	1.0 (1.0)	2.3 (2.2)	1.6 (1.9)
N	287	458	384	113	516	347

predictors of cultural access, while immigration and residential mobility are negatively associated with it. These findings suggest that socioeconomic and mobility factors significantly mediate cultural access in Black and Hispanic neighborhoods, but have less effect on Asian neighborhoods. Building on the descriptive findings of neighborhood typology, we argue that the predictive influence of socioeconomic and mobility factors is closely tied to the socioeconomic and mobility characteristics of these neighborhoods; in other words, their significance suggests the access is stratified based on the different ethnoracial neighborhood types. On the other hand, concentration immigration remains negatively significant, indicating reduced cultural access in general.

To further investigate how immigration intersects with ethnoracial composition, Models 3 and 4 add interaction terms between neighborhood type and concentrated immigration. It is worth noting that the predominantly minority immigrant neighborhoods have greater access than their nonimmigrant counterparts. Specifically, Figure 4 presents the marginal effects for the interaction of neighborhood types and concentrated immigration in Model 4. Immigrant neighborhoods have divergent amounts of access for different neighborhood types: for White neighborhoods and non-White mixed neighborhoods, the amount of access sharply declines by concentrated immigration; in contrast, for Hispanic and Asian neighborhoods, cultural accessibility increases by immigration. Black neighborhoods also have a slightly increased amount of access due to immigration, but the increase is not apparent.

We also analyze these relationships specifically for ethnic-oriented and citywide organizations. As shown in Table 4, the findings remain consistent across different types of organizations, though there are variations in the scale of access. Asian neighborhoods demonstrate the highest access to both ethnic-oriented citywide organizations, followed by White mixed neighborhoods, while Black neighborhoods have the least access. However, the disparity in access to ethnic-oriented

Table 3. Regressions models for number of cultural organizations within 1 mile.

	All organizations			
	Model 1	Model 2	Model 3	Model 4
Predominantly Black	−22.7***	17.6***	−15.1**	28.0***
	(4.3)	(4.0)	(5.7)	(5.1)
Predominantly Hispanic	−2.4	30.0***	−9.0	25.4***
	(4.9)	(4.6)	(5.6)	(5.1)
Predominantly Asian	33.5***	34.2***	2.0	8.9
	(7.6)	(6.5)	(14.2)	(12.2)
White mixed	22.5***	10.0**	22.7***	15.0***
	(4.3)	(3.7)	(5.1)	(4.4)
Non-White mixed	5.9	23.8***	9.2	29.9***
	(4.9)	(4.3)	(5.5)	(4.8)
Concentrated immigration	−16.5***	−6.6***	−21.0***	−15.1***
	(1.2)	(1.1)	(3.0)	(2.6)
% Residents in poverty		1.3***		1.4***
		(0.1)		(0.1)
% Residents with college degree		2.5***		2.5***
		(0.1)		(0.1)
County-level mobility		−1.5*		−1.7**
		(0.6)		(0.6)
Black x Immigration			10.2*	16.4***
			(5.2)	(4.5)
Hispanic x Immigration			21.2***	22.3***
			(4.1)	(3.5)
Asian x Immigration			19.3**	21.4***
			(6.0)	(5.1)
White mixed x Immigration			−9.9**	−2.6
			(3.7)	(3.2)
Non-White mixed x Immigration			6.8	9.3**
			(4.0)	(3.5)
Constant	28.3***	−124.1***	23.9***	−131.2***
	(3.6)	(6.3)	(4.4)	(6.8)
R-squared	0.148	0.385	0.185	0.411
Adjusted R-squared	0.145	0.382	0.181	0.407
N	2,105	2,105	2,105	2,105

*$p < .05$; **$p < .01$; ***$p < .001$; Reference group: predominantly White.

organizations for Black neighborhoods has significantly diminished, which suggests a greater degree of racial segregation in access to citywide resources than ethnic resources for Black neighborhoods.

Similarly, we visualize the interaction between neighborhood type and immigration for number of ethnic-oriented and citywide organizations within 1 mile, presented in Figure 5 and 6 respectively. Consistent with previous findings, access to ethnic-oriented organizations increases with immigration in Hispanic and Asian neighborhoods. Notably, Hispanic neighborhoods exhibit a steeper slope, indicating that greater access is more strongly associated with immigrant concentration. Meanwhile, access to ethnic-oriented organizations decreases with immigration for other ethnoracial neighborhoods, with sharper rates observed in White neighborhoods and more moderate rates in Black and non-White mixed neighborhoods. In contrast, the patterns of access to citywide organizations align more closely with the overall trends in cultural access. Black neighborhoods also show increased access with higher levels of immigration, after controlling for socioeconomic and mobility factors. Overall, these findings highlight a consistent relationship for overall cultural access and access specific to different organization types.

Next, we investigate the patterns for distance to nearest cultural organization, as shown in Table 5 (all organizations) and Table 6 (ethnic-oriented and citywide organizations).

For the overall distance models, Hispanic and White mixed neighborhoods consistently have shorter distance to organizations than predominantly White neighborhoods. Black and immigrant neighborhoods have greater access when controlling for socioeconomic and mobility factors,

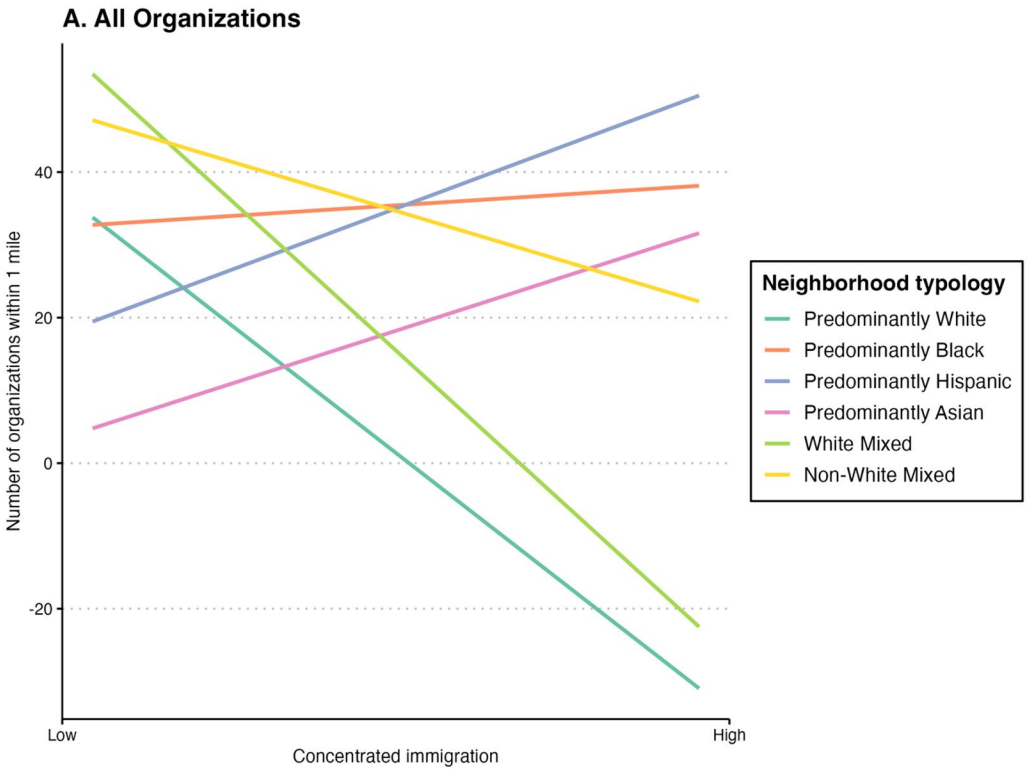


Figure 4. Number of cultural organizations within 1 mile by neighborhood typology and immigration level.

while Asian and non-White mixed neighborhoods are not significant. On the other hand, all three control variables are negative predictors for accessibility.

Since count and distance measure reflects different aspects of cultural access, the differing effects of socioeconomic and mobility factors between the two outcomes suggest that the distance measure is more sensitive to a neighborhood's location. Specifically, high poverty rates, high levels of education and high mobility rates are closely associated with the characteristics of urban centers. In other words, the distance models indicate that Hispanic and White mixed neighborhoods are more likely to be located in or near urban centers compared to other neighborhoods. Additionally, while the results for concentrated immigration in the distance models may initially appear to contradict those in the count models, the interaction between neighborhood types and immigration helps clarify this discrepancy.

Figure 7 illustrates the marginal effects for this interaction, showing that Hispanic and Asian neighborhoods experience greater cultural accessibility with higher levels of concentrated immigration. Moreover, immigrant neighborhoods in these areas have the shortest distances to cultural resources compared to other neighborhood types. Distance to nearest organization also decline by immigration level for other types of neighborhoods, except for Black neighborhoods, which experience slightly reduced cultural accessibility by immigration. Combined with previous findings, these patterns indicate that the overall increase in cultural accessibility associated with immigration is largely driven by Hispanic and Asian immigrant neighborhoods.

Table 6 displays the coefficients from the distance models for both ethnic-oriented and citywide organizations. For ethnic-oriented organizations, all neighborhood types as well as immigrant neighborhoods have significantly shorter distance compared to predominantly White neighborhoods, whereas the patterns for citywide organizations are consistent with those observed for overall access. It is worth noting that Asian neighborhoods have the shortest distances to ethnic-oriented organizations, while Hispanic and White mixed neighborhoods are closest to

Table 4. Regression models for ethnic-oriented and citywide organizations within 1 mile.

	Ethnic		Citywide	
	Model 1	Model 2	Model 3	Model 4
Predominantly Black	−0.8**	1.9***	−10.8***	11.2***
	(0.3)	(0.3)	(1.8)	(2.2)
Predominantly Hispanic	0.3	1.9***	−3.2	10.3***
	(0.3)	(0.3)	(2.1)	(2.2)
Predominantly Asian	2.7***	1.5	10.4**	0.8
	(0.5)	(0.8)	(3.2)	(5.2)
White mixed	1.3***	1.1***	8.6***	6.3***
	(0.3)	(0.3)	(1.8)	(1.9)
Non-White mixed	0.9**	2.4***	0.6	11.2***
	(0.3)	(0.3)	(2.1)	(2.0)
Concentrated immigration	−0.8***	−0.8***	−6.3***	−6.1***
	(0.1)	(0.2)	(0.5)	(1.1)
% Residents in poverty		0.1***		0.5***
		(0.01)		(0.04)
% Residents with college degree		0.1***		1.0***
		(0.01)		(0.04)
County-level mobility		−0.2***		−0.9***
		(0.04)		(0.3)
Black x Immigration		0.7*		7.7***
		(0.3)		(1.9)
Hispanic x Immigration		1.6***		8.7***
		(0.2)		(1.5)
Asian x Immigration		1.3***		9.3***
		(0.3)		(2.2)
White mixed x Immigration		−0.03		−0.2
		(0.2)		(1.4)
Non-White mixed x Immigration		0.5*		4.3**
		(0.2)		(1.5)
Constant	1.6***	−7.2***	11.5***	−53.7***
	(0.2)	(0.5)	(1.5)	(2.9)
R-squared	0.081	0.301	0.142	0.391
Adjusted R-squared	0.079	0.296	0.140	0.387
N	2,105	2,105	2,105	2,105

*$p < .05$; **$p < .01$; ***$p < .001$; Reference group: predominantly White.

citywide organizations, aligning with previous findings from count models. Figure 8 further illustrate the "distance advantage" associated with immigration levels for ethnic-oriented organizations, and Figure 9 confirms similar interaction patterns for citywide organizations as overall organizations.

The two cultural accessibility measures tell the story in different ways: White mixed neighborhoods consistently enjoy greater cultural accessibility, while Black neighborhoods are disadvantaged in access. Ethnic-oriented organizations primarily benefit minority and immigrant neighborhoods, whereas citywide organizations mirror the overall patterns. When immigration comes into play, the divergent patterns emerge by race and ethnicity. The overall immigration effect easily masks the positive relationship for Hispanic and Asian immigrant neighborhoods. Hispanic and Asian neighborhoods enjoy better access with greater immigration level, while the White and Black immigrant neighborhoods do not share the same advantages. In fact, Hispanic immigrant enclaves have the best cultural accessibility among all, followed by Asian immigrant enclaves. For Black neighborhoods, however, the cultural accessibility patterns are mainly explained by socioeconomic and mobility factors, while immigration plays a limited role.

To summarize our findings, the interaction between immigration and ethnoracial neighborhoods lays out different trajectories of cultural accessibility: higher immigrant concentration reduces overall access in White neighborhoods, but in Hispanic and Asian neighborhoods, it increases access. Moreover, immigrant concentration largely increases access to ethnic-oriented

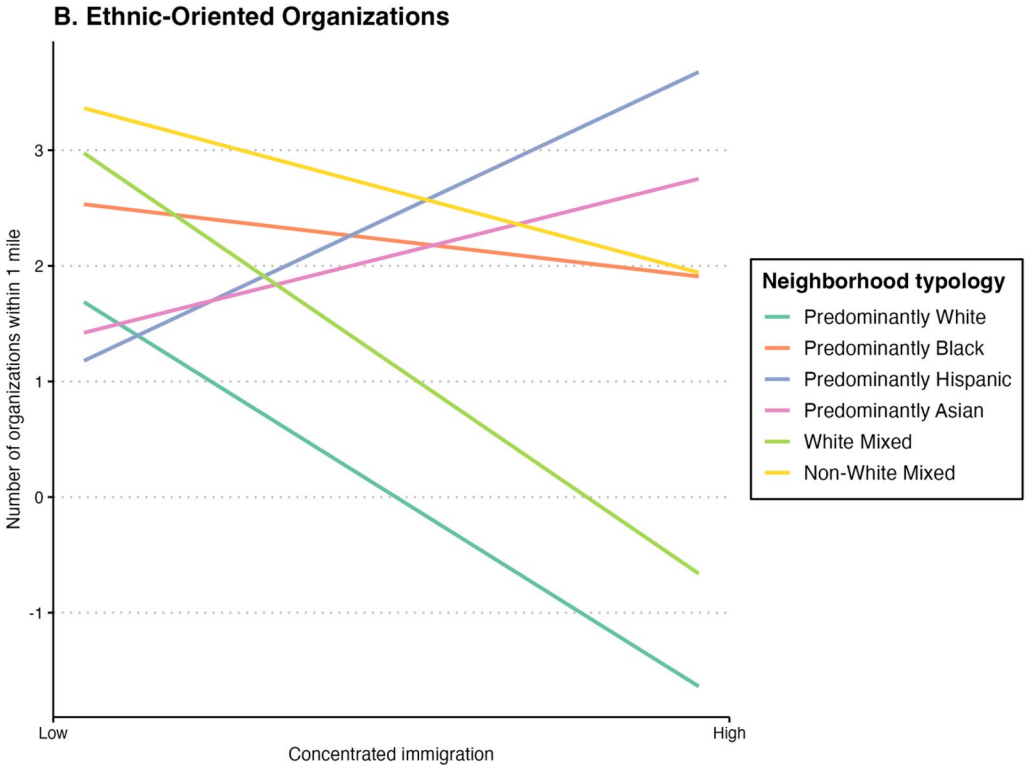


Figure 5. Number of ethnic-oriented cultural organizations within 1 mile by neighborhood typology and immigration level.

organizations, mainly by reducing walking distance. For Black neighborhoods, immigration plays a minimal role in access, with socioeconomic and mobility factors reducing their access.

Conclusion and discussion

In this study, we applied spatial techniques to quantify cultural accessibility in two ways: we measured the number of cultural organizations within 1 mile of neighborhood surroundings, and we also calculated the distance to the nearest cultural organization for each neighborhood. Using these two measures, we analyzed how cultural accessibility differ by neighborhood's ethnoracial types and concentrated immigration level, with socioeconomic and residential mobility factors considered. We observed divergent patterns at the intersection of neighborhood types and immigration, with five key takeaways from our findings. First, the degree of cultural accessibility for different neighborhood types varies according to their immigrant concentration. For Hispanic and Asian neighborhoods, concentrated immigration is associated with greater cultural accessibility, but this relationship is not always significant for other minority neighborhood types. Moreover, White neighborhoods show reverse patterns due to concentrated immigration, in that those with a higher concentration of immigrants have lower cultural accessibility. This indicates that immigrant neighborhoods are not always worse off: the degree of cultural accessibility for immigrant neighborhoods also differs by the composition of race and ethnicity in those neighborhoods.

Second, our results also show that Hispanic and Asian immigrant neighborhoods have the highest level cultural accessibility among all. There are two reasons behind this phenomenon. As Stern et al. (2010) mentioned in their study of Philadelphia's cultural economy, Hispanic and Asian immigrant neighborhoods are usually located in or near the city center. In New York City, this also comes true. Hispanic and Asian immigrants of recent waves tend to reside in

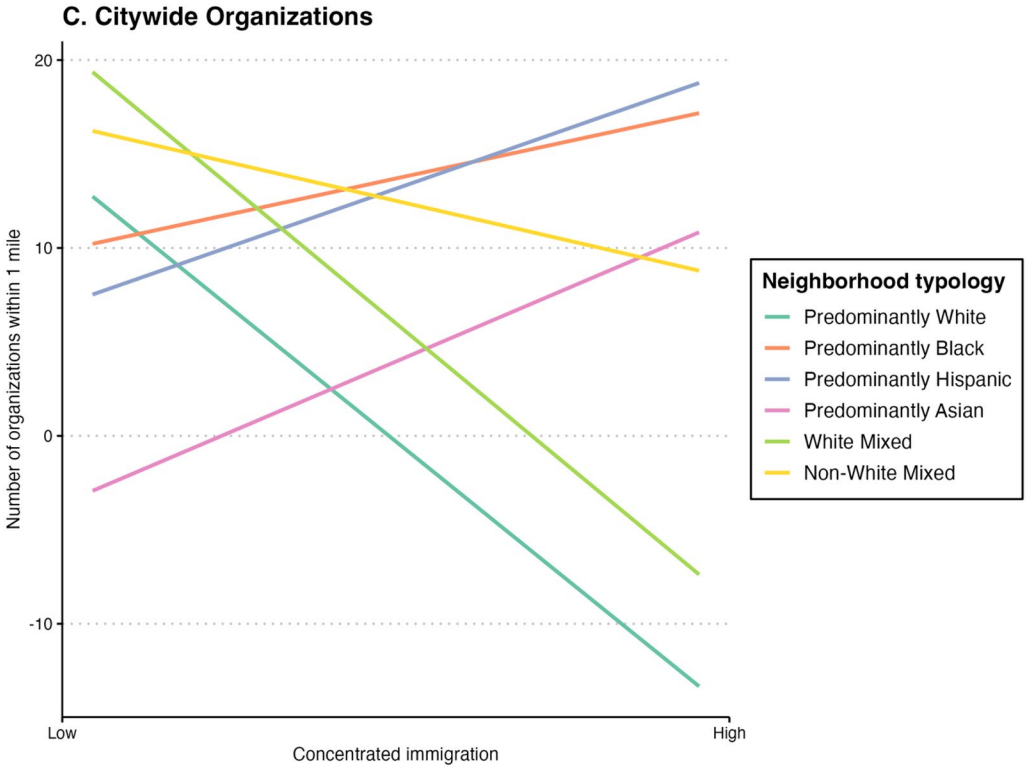


Figure 6. Number of citywide cultural organizations within 1 mile by neighborhood typology and immigration level.

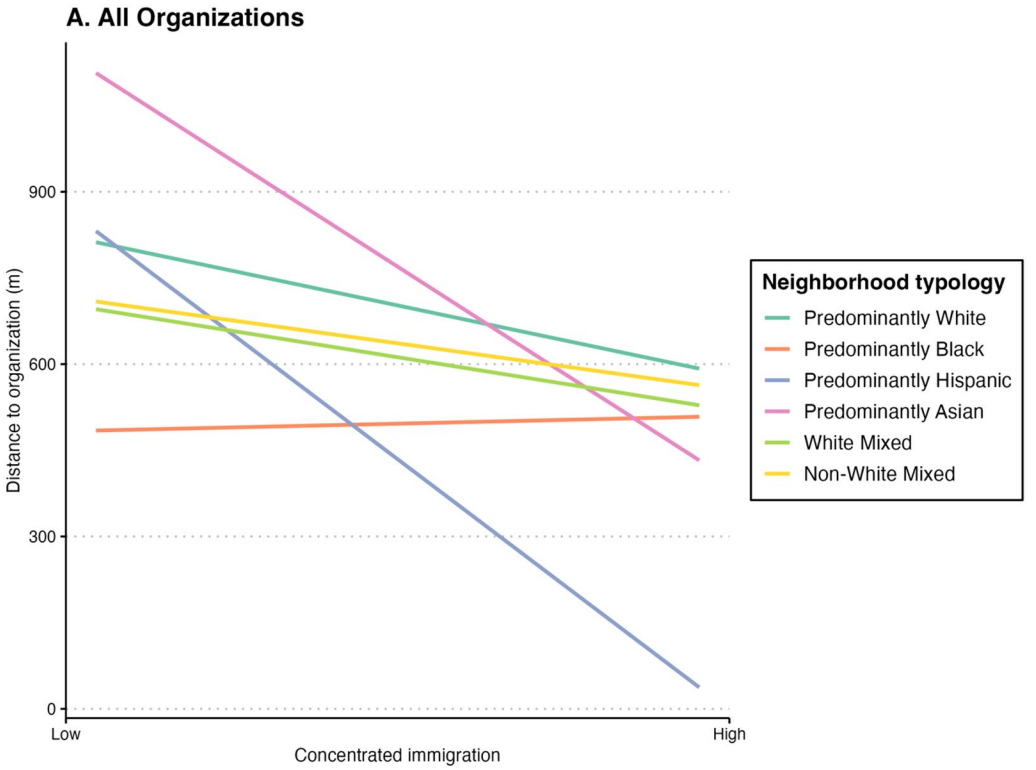


Figure 7. Distance to nearest cultural organizations by neighborhood typology and immigration.

Table 5. Regressions models for distance to nearest cultural organization.

	All organizations			
	Model 1	Model 2	Model 3	Model 4
Predominantly Black	−8.1	−254.2***	73.8	−226.7***
	(35.5)	(34.0)	(46.6)	(43.2)
Predominantly Hispanic	−144.5***	−275.3***	−45.1	−218.6***
	(40.0)	(39.1)	(45.9)	(43.9)
Predominantly Asian	−115.8	−80.4	199.3	106.3
	(62.1)	(55.0)	(117.3)	(103.9)
White mixed	−190.0***	−81.1**	−156.6***	−94.7*
	(35.1)	(31.4)	(42.2)	(37.5)
Non-White mixed	44.5	−36.1	44.6	−72.3
	(40.0)	(36.1)	(45.3)	(40.7)
Concentrated immigration	−2.9	−73.2***	−20.8	−51.3*
	(10.0)	(9.4)	(25.0)	(22.6)
% Residents in poverty		−12.1***		−11.9***
		(0.8)		(0.8)
% Residents with college degree		−16.8***		−16.6***
		(0.8)		(0.7)
County-level mobility		−17.1***		−14.8**
		(5.2)		(5.1)
Black x Immigration			134.0**	56.9
			(42.7)	(38.8)
Hispanic x Immigration			−107.8**	−134.0***
			(33.5)	(30.0)
Asian x Immigration			−105.3*	−105.9*
			(49.3)	(43.6)
White mixed x Immigration			73.1*	12.3
			(30.8)	(27.3)
Non-White mixed x Immigration			55.7	17.4
			(33.1)	(29.5)
Constant	640.2***	1,762.4***	622.7***	1,764.3***
	(29.3)	(53.7)	(36.6)	(58.4)
R-squared	0.037	0.248	0.066	0.267
Adjusted R-squared	0.034	0.245	0.061	0.262
N	2,105	2,105	2,105	2,105

*$p < .05$; **$p < .01$; ***$p < .001$; Reference group: predominantly White.

areas with a longer immigration tradition established by early European immigrant waves. In addition, the higher concentration of Hispanic and Asian immigrants within these neighborhoods also benefits them with strong migrant network ties, which may contribute to their better access to cultural resources. The institutional infrastructure and ethnic solidarity play a role in neighborhoods with high concentrations of Hispanic and Asian immigrants are consistent with their greater cultural accessibility.

Third, as minority neighborhoods, Black neighborhoods exhibit different patterns from Hispanic and Asian neighborhoods. The lower cultural accessibility of Black neighborhoods is mainly explained by the socioeconomic status and residential mobility of these neighborhoods. Immigration has little effect on the pattern. This suggests that socioeconomic disadvantage and residential segregation mainly factor into low cultural accessibility for Black neighborhoods, which reaffirms the long-term effect of racial segregation for Black poverty (Massey and Denton 1988). Prior research has extensively documented that the socioeconomic disadvantage observed in Black neighborhoods is rooted in a long history of systematic racism and institutional disinvestment, as evidenced by decades of residential segregation, redlining and discriminatory lending practices (Massey and Denton 1988; Rothstein 2018; Sharkey 2013). These patterns of exclusion also extend to the broader institutional landscape, and our finding regarding the lower cultural accessibility of Black neighborhoods is consistent with this broad body of research.

Fourth, it is worth noting that White mixed neighborhoods also have better access than predominantly White neighborhoods. Non-White mixed neighborhoods also perform better if

Table 6. Regressions models for distance to nearest ethnic-oriented and citywide organization.

	Ethnic-oriented		Citywide	
	Model 1	Model 2	Model 3	Model 4
Predominantly Black	−165.2	−771.7***	−43.4	−389.1***
	(98.0)	(124.4)	(59.1)	(72.5)
Predominantly Hispanic	−781.4***	−1,052.8***	−328.5***	−490.4***
	(110.4)	(126.5)	(66.6)	(73.7)
Predominantly Asian	−1,044.8***	−770.5*	−274.5**	232.1
	(171.4)	(299.5)	(103.4)	(174.5)
White mixed	−817.9***	−589.3***	−340.1***	−190.3**
	(97.0)	(108.1)	(58.5)	(63.0)
Non-White mixed	−765.5***	−1,029.4***	−37.2	−207.5**
	(110.5)	(117.3)	(66.7)	(68.3)
Concentrated immigration	−37.9	−175.2**	23.5	−46.8
	(27.6)	(65.0)	(16.7)	(37.9)
% Residents in poverty		−31.5***		−20.3***
		(2.2)		(1.3)
% Residents with college degree		−42.7***		−27.6***
		(2.2)		(1.3)
County-level mobility		3.0		−22.3**
		(14.8)		(8.6)
Black x Immigration		−3.6		113.5
		(111.9)		(65.2)
Hispanic x Immigration		−163.9		−176.3***
		(86.5)		(50.4)
Asian x Immigration		−116.4		−244.8***
		(125.6)		(73.2)
White mixed x Immigration		30.3		17.5
		(78.7)		(45.9)
Non-White mixed x Immigration		22.6		−48.0
		(85.1)		(49.6)
Constant	2,197.9***	5,024.1***	1,162.6***	3,046.9***
	(80.9)	(168.3)	(48.8)	(98.1)
R-squared	0.077	0.241	0.034	0.259
Adjusted R-squared	0.075	0.236	0.031	0.254
N	2,105	2,105	2,105	2,105

*$p < .05$; **$p < .01$; ***$p < .001$; Reference group: predominantly White.

controlling for socioeconomic and mobility factors. The advantage of cultural accessibility among mixed neighborhoods with a certain percentage of White groups suggests that ethnic diversity is positively associated with access to cultural resources.

Finally, ethnic-oriented organizations enhance cultural access across immigrant neighborhoods of diverse ethnoracial compositions. In contrast, citywide organizations remain spatially stratified based on neighborhood locations. The patterns of citywide organizations mirror those of overall cultural access, indicating that the citywide organizations play a significant role in shaping overall cultural accessibility. This highlights the persistent spatial inequalities in access to cultural resources.

Overall, we discussed the cultural accessibility in New York City and highlighted how it differs by neighborhood's ethnoracial composition and concentrated immigration. To some extent, our findings on the greater cultural accessibility in Asian and Hispanic immigrant neighborhoods echo the idea that Asians and Hispanic neighborhoods have distinctive trajectories from Black neighborhoods given their large-scale immigration and lower level of segregation (Alba 2024; Logan and Stults 2021). This suggests that cultural access may represent a unique dimension of the assimilation and integration experience for Hispanic and Asian immigrants, given the proximity to ethnic resources and locations of their neighborhoods.

Our study is not without limitations. First, our analyses focus exclusively on nonprofit cultural organizations as recorded in the DCLA directory. We acknowledge that for-profit and informal cultural organizations as well as digital platforms can supplement cultural accessibility, allowing

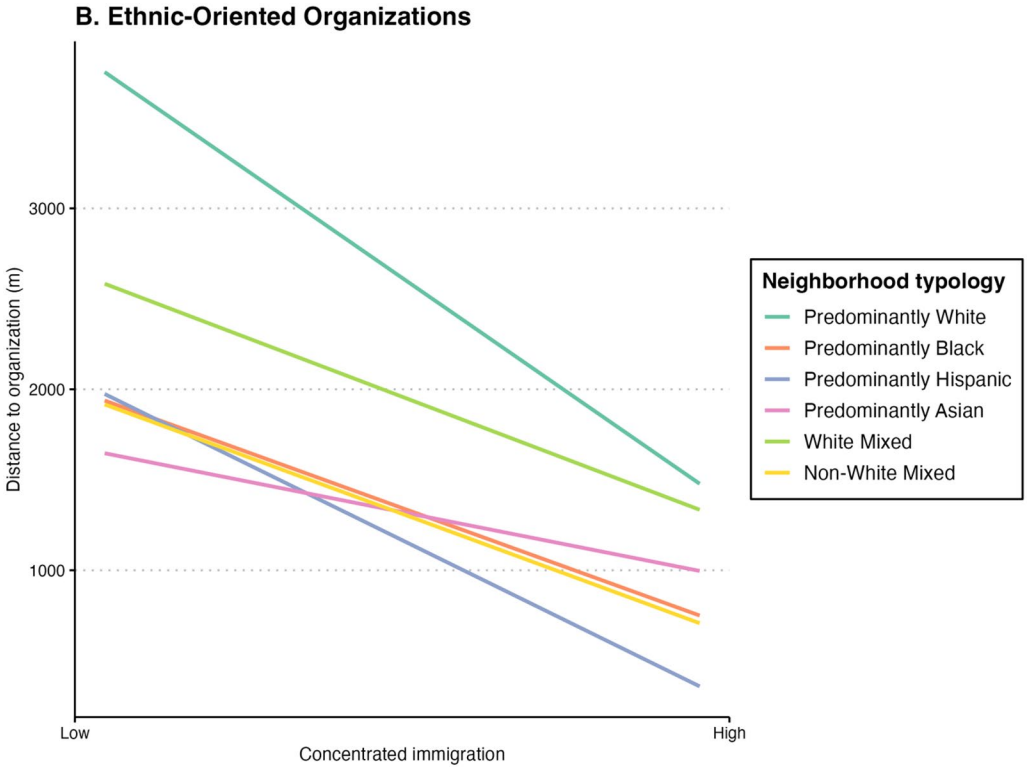


Figure 8. Distance to nearest ethnic-oriented cultural organizations by neighborhood typology and immigration.

people to access cultural resources beyond their local neighborhoods. People are also likely to travel across neighborhoods for cultural access, especially to renowned institutions like the Met Museum and MoMA. However, we argue that the neighborhood context remains crucial for meeting the routine cultural needs of local communities, especially for ethnoracial minorities and immigrants, who are more likely to face limited resources and mobility in accessing services beyond their neighborhoods. In this regard, nonprofit cultural organizations play a particularly vital role in serving local communities, as they typically offer free or discounted events and programs, making culture more accessible to diverse audiences.

Our finding that neighborhoods of ethnoracial with higher immigrant concentration are closer to ethnic-oriented organizations further supports this point. Second, our measure of cultural accessibility captures potential access to cultural organizations rather than actual participation or engagement. We acknowledge that our measures reflect the opportunity structure available to neighborhood residents, while they do not offer insights on observed patterns of cultural attendance or program use. Nevertheless, spatial accessibility provides a meaningful baseline for understanding inequality in the distribution of cultural resources, and future research may build on this work by incorporating data on actual participation. Third, we use Euclidean distance to measure spatial proximity between census tract centroids and cultural organizations. We adopt this approach because New York City presents a particularly suitable urban context given its dense and grid-patterned street network and high walkability.

In a metropolitan context, the Euclidean distance and network distance are highly correlated, so the relative rankings of spatial proximity are stable under either metric (Apparicio et al. 2008). That said, we acknowledge that network-based distance measures would more precisely capture actual travel routes, and future research could employ network distance or travel-time measures to refine estimates of cultural accessibility. Fourth, due to data limitations, we are

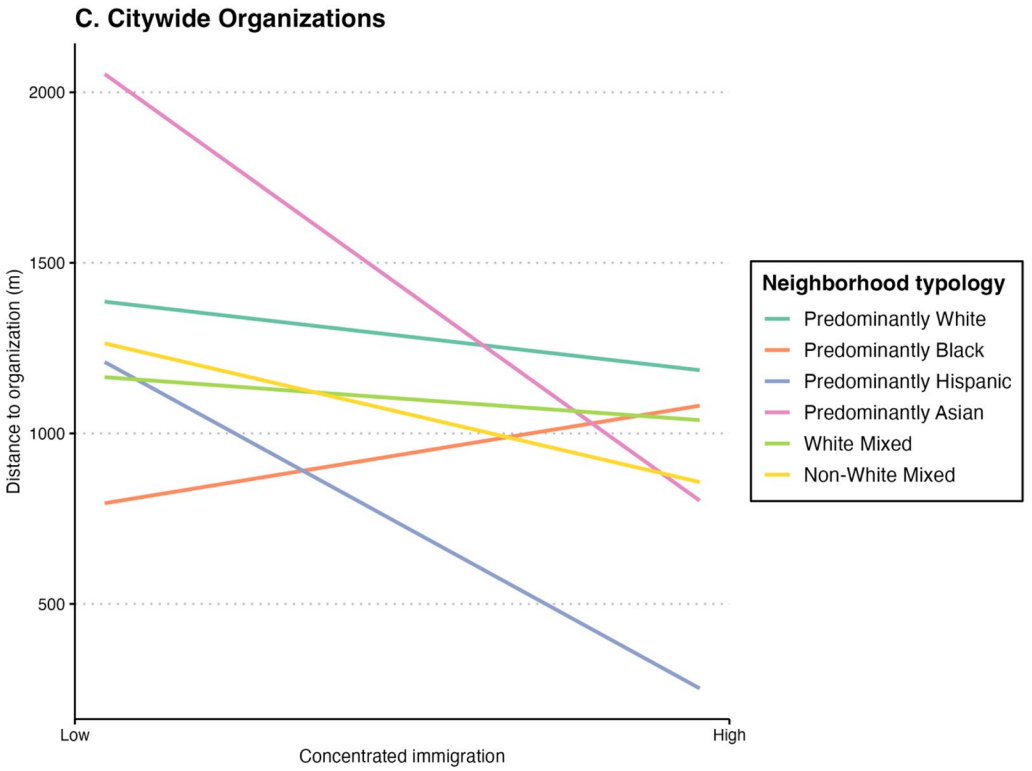


Figure 9. Distance to nearest citywide cultural organizations by neighborhood typology and immigration.

unable to investigate the temporal dynamics of cultural organizations across neighborhoods because the data we used comes from a directory that lacks multiyear variations. This limits our ability to draw causal conclusions about the relationship between the spatial distribution of cultural organizations and neighborhood characteristics. Our findings should therefore be interpreted as association rather than causation. Nonetheless, we contend that the distinct patterns of cultural accessibility observed in our findings, particularly the divergent trajectories of Hispanic and Asian neighborhoods compared to Black neighborhoods, are consistent with the broader neighborhood patterns documented in the urban and immigration literature, which are largely shaped by mass immigration.

Our study addresses the connections between urban settings and cultural networks, and the crucial role of cultural access in immigrant assimilation and integration. As our findings indicate that neighborhoods with higher immigrant concentrations tend to have better cultural accessibility, we argue that cultural access may play an important role in immigrant assimilation and integration. Policymakers should therefore consider incorporating nonprofit cultural organizations into neighborhood revitalization efforts, particularly in the Black neighborhoods. The city should also ensure cultural funding in both socioeconomically disadvantaged and segregated neighborhoods, to foster their access to social capital among residents. Both policymakers and scholars should also address the issue of segregation from a cultural perspective, integrating these efforts into housing policies, education initiatives and community-oriented programs. Future research should investigate the extent to which cultural accessibility of a neighborhood may affect individual life course in the long run, such as career choice and job market performance. It is also important to consider how the cultural aspects of organizational infrastructure are associated with neighborhood outcomes, especially for neighborhoods with a high concentration of minority groups and immigrants. We hope that our work will help advocate for a further consideration of equal spatial access to cultural resources.

Author contributions

CRediT: **Qiyao Pan**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Juliette Zhu**: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research was supported by the Dr. Louise Lennihan Arts and Sciences Grant from the Futures Initiative at the Graduate Center, City University of New York.

Author Notes

Qiyao Pan is a Ph.D. candidate in Sociology at the Graduate Center, CUNY. Her research interests are broadly in urban sociology, immigration, spatial demography and computational social science. Her current research mainly examines the determinants of neighborhood services, neighborhood dynamics and migration patterns and using spatial and computational analysis.

Juliette Zhu received a Ph.D. in Sociology from the Graduate Center, CUNY. Her research focuses on contemporary family life. She uses both qualitative and quantitative research methods to study topics include global migration, education, gender, and animal-human relationships.

ORCID

Qiyao Pan http://orcid.org/0009-0006-4744-6263
Juliette Zhu http://orcid.org/0000-0002-4746-5521

References

Alba, Richard. 2024. "Culture's Role in Assimilation and Integration: The Expansion and Growing Diversity of U.S. Popular Culture." *Journal of Ethnic and Migration Studies 50* (1):27–46. doi: 10.1080/1369183X.2023.2213046.

Apparicio, Philippe, Mohamed Abdelmajid, Mylène Riva, and Richard Shearmur. 2008. "Comparing Alternative Approaches to Measuring the Geographical Accessibility of Urban Health Services: Distance Types and Aggregation-Error Issues." *International Journal of Health Geographics 7* (1):7. doi: 10.1186/1476-072X-7-7.

Borer, Michael Ian. 2006. "The Location of Culture: The Urban Culturalist Perspective." *City & Community 5* (2):173–97. doi: 10.1111/j.1540-6040.2006.00168.x.

Bourdieu, Pierre. 1986. "The Forms of Capital." Pp. 241–58 in *Handbook of Theory and Research for the Sociology of Education*. Westport, CT: Greenwood.

Bridge, Gary. 2006. "Perspectives on Cultural Capital and the Neighbourhood." *Urban Studies 43* (4):719–30. doi: 10.1080/00420980600597392.

Bruno, Matteo, Hygor Piaget Monteiro Melo, Bruno Campanelli, and Vittorio Loreto. 2024. "A Universal Framework for Inclusive 15-Minute Cities." *Nature Cities 1* (10):633–41. doi: 10.1038/s44284-024-00119-4.

Cheng, Yuan (Daniel), Lang (Kate) Yang, and Shuyi Deng. 2022. "Nonprofit Density and Distributional Equity in Public Service Provision: Exploring Racial/Ethnic Disparities in Public Park Access across U.S. Cities." *Public Administration Review 82* (3):473–86. doi: 10.1111/puar.13465.

Clark, Terry Nichols. 2003. *The City as an Entertainment Machine*. UK: Emerald Group Publishing Limited.

Coleman, James S. 1988. "Social Capital in the Creation of Human Capital." *American Journal of Sociology 94*:S95–S120. doi: 10.1086/228943.

Curley, Alexandra M. 2010. "Relocating the Poor: Social Capital and Neighborhood Resources." *Journal of Urban Affairs 32* (1):79–103. doi: 10.1111/j.1467-9906.2009.00475.x.

Currid-Halkett, Elizabeth. 2007. *The Warhol Economy: How Fashion, Art, and Music Drive New York City*. Princeton, NJ: Princeton University Press.
DiMaggio, Paul, and Patricia Fernandes-Kelly, eds. 2010. *Art in the Lives of Immigrant Communities in the United States*. New Brunswick, NJ: Rutgers University Press.
Downey, Liam. 2003. "Spatial Measurement, Geography, and Urban Racial Inequality*." *Social Forces 81* (3):937–52. doi: 10.1353/sof.2003.0031.
Dreier, Peter, John Mollenkopf, and Todd Swanstrom. 2014. Place Matters: Metropolitics for the Twenty-First Century. 3rd ed., Revised. Lawrence, KS: University Press of Kansas.
Edwards, B., and J. D. McCarthy. 2004. "Strategy Matters: The Contingent Value of Social Capital in the Survival of Local Social Movement Organizations*." *Social Forces 83* (2):621–51. doi: 10.1353/sof.2005.0009.
Elliott, James R., Elizabeth Korver–Glenn, and Daniel Bolger. 2019. "The Successive Nature of City Parks: Making and Remaking Unequal Access over Time." *City & Community 18* (1):109–27. doi: 10.1111/cico.12366.
Farrell, Chad R., and Barrett A. Lee. 2011. "Racial Diversity and Change in Metropolitan Neighborhoods." *Social Science Research 40* (4):1108–23. doi: 10.1016/j.ssresearch.2011.04.003.
Fernández-Kelly, Patricia. 2015. *The Hero's Fight: African Americans in West Baltimore and the Shadow of the State*. Princeton, NJ: Princeton University Press.
Florida, Richard. 2003. "Cities and the Creative Class." *City & Community 2* (1):3–19. doi: 10.1111/1540-6040.00034.
Foner, Nancy. 2023. *One Quarter of the Nation: Immigration and the Transformation of America*. Princeton, NJ: Princeton University Press.
Foster, Kirk, A. Ronald Pitner, Darcy A. Freedman, Bethany A. Bell, and Todd C. Shaw., 2015. "Spatial Dimensions of Social Capital." *City & Community 14* (4):392–409. doi: 10.1111/cico.12133.
Fullilove, Mindy Thompson. 2016. *Root Shock: How Tearing Up City Neighborhoods Hurts America, And What We Can Do About It*. New York: New Village Press.
Galster, George, and Patrick Sharkey. 2017. "Spatial Foundations of Inequality: A Conceptual Model and Empirical Overview." *RSF: The Russell Sage Foundation Journal of the Social Sciences 3* (2):1–33. doi: 10.7758/rsf.2017.3.2.01.
Graif, Corina. 2018. "Neighborhood Diversity and the Rise of Artist Hotspots: Exploring the Creative Class Thesis Through a Neighborhood Change Lens." *City & Community 17* (3):754–87. doi: 10.1111/cico.12317.
Grazian, David. 2008. *On the Make: The Hustle of Urban Nightlife*. Chicago: University of Chicago Press.
Hall, Matthew, Kyle Crowder, and Amy Spring. 2015. "Neighborhood Foreclosures, Racial/Ethnic Transitions, and Residential Segregation." *American Sociological Review 80* (3):526–49. doi: 10.1177/0003122415581334.
Hwang, Hyunseok, and Young-Joo Lee. 2023. "Community Social Capital, Racial Diversity, and Philanthropic Resource Mobilization in the Time of a Pandemic." *City & Community 22* (1):22–47. doi: 10.1177/15356841221119181.
Jorgensen, Bradley S., and Richard C. Stedman. 2011. "Measuring the Spatial Component of Sense of Place: A Methodology for Research on the Spatial Dynamics of Psychological Experiences of Places." *Environment and Planning B: Planning and Design 38* (5):795–813. doi: 10.1068/b37054.
Kasinitz, Philip. 2023. "Global Commerce, Immigration and Diversity: A New York Story." *Ethnic and Racial Studies 46* (11):2533–54. doi: 10.1080/01419870.2023.2174808.
Kasinitz, Philip, and Marco Martiniello. 2019. "Music, Migration and the City." *Ethnic and Racial Studies 42* (6):857–64. doi: 10.1080/01419870.2019.1567930.
Kasinitz Philip, John H. Mollenkopf, Mary C. Waters, and Jennifer Holdaway. 2009. Inheriting the *city*: The *children of immigrants come of age*. New York: Russell Sage Foundation.
Kawlra, Gayatri, and Kazuki Sakamoto. 2023. "Spatialising Urban Health Vulnerability: An Analysis of NYC's Critical Infrastructure during COVID-19." *Urban Studies (Edinburgh, Scotland) 60* (9):1629–49. doi: 10.1177/00420980211044304.
Kennedy, Ian, Chris Hess, Amandalynne Paullada, and Sarah Chasins. 2021. "Racialized Discourse in Seattle Rental Ad Texts." *Social Forces; a Scientific Medium of Social Study and Interpretation 99* (4):1432–56. doi: 10.1093/sf/soaa075.
Lizardo, Omar, and Isaac Jilbert. 2023. "Organizations and the Structure of Culture." *Sociology Compass 17* (4):e13063. doi: 10.1111/soc4.13063.
Lloyd, Richard. 2004. "The Neighborhood in Cultural Production: Material and Symbolic Resources in the New Bohemia." *City & Community 3* (4):343–72. doi: 10.1111/j.1535-6841.2004.00092.x.
Lloyd, Richard. 2010. *Neo-Bohemia: Art and Commerce in the Postindustrial City*. New York: Routledge.
Logan, John R. 2012. "Making a Place for Space: Spatial Thinking in Social Science." *Annual Review of Sociology 38* (1):507–24. doi: 10.1146/annurev-soc-071811-145531.
Logan, John R., Seth Spielman, Hongwei Xu, and Philip N. Klein. 2011. "Identifying and Bounding Ethnic Neighborhoods." *Urban Geography 32* (3):334–59. doi: 10.2747/0272-3638.32.3.334.
Logan, John R., and Brian Stults. 2021. "Metropolitan Segregation: No Breakthrough in Sight." Brown University. https://s4.ad.brown.edu/Projects/Diversity/Data/Report/report08122021.pdf
Massey, Douglas S., and Nancy A. Denton. 1988. "The Dimensions of Residential Segregation." *Social Forces 67* (2):281–315. doi: 10.2307/2579183.

McKenzie, Brian S. 2013. "Neighborhood Access to Transit by Race, Ethnicity, and Poverty in Portland, OR." *City & Community 12* (2):134–55. doi: 10.1111/cico.12022.
Ocejo, Richard E. 2017. *Upscaling Downtown: From Bowery Saloons to Cocktail Bars in New York City*. Princeton, NJ: Princeton University Press.
Rhubart, Danielle, Yue Sun, Claire Pendergrast, and Shannon Monnat. 2022. "Sociospatial Disparities in 'Third Place' Availability in the United States." *Socius: sociological Research for a Dynamic World 8*:23780231221090301. doi: 10.1177/23780231221090301.
Rothstein, Richard. 2018. *The Color of Law: A Forgotten History of How Our Government Segregated America*. New York: Liveright Publishing Corporation
Roubenoff, Ethan, Jasmijn Slootjes, and Irene Bloemraad. 2023. "Spatial and Sociodemographic Vulnerability: Quantifying Accessibility to Health Care and Legal Services for Immigrants in California, Arizona, and Nevada." *Socius: Sociological Research for a Dynamic World 9*:23780231231157683. doi: 10.1177/23780231231157683.
Ruef, Martin, and Seok-Woo Kwon. 2016. "Neighborhood Associations and Social Capital." *Social Forces 95* (1):159–90. doi: 10.1093/sf/sow053.
Rutten, Roel, Hans Westlund, and Frans Boekema. 2010. "The Spatial Dimension of Social Capital." *European Planning Studies 18* (6):863–71. doi: 10.1080/09654311003701381.
Sampson, R. J., S. W. Raudenbush, and F. Earls. 1997. "Neighborhoods and Violent Crime: A Multilevel Study of Collective Efficacy." *Science (New York, N.Y.) 277* (5328):918–24. doi: 10.1126/science.277.5328.918.
Sampson, Robert J. 2012. *Great American City: Chicago and the Enduring Neighborhood Effect*. Chicago: University of Chicago Press.
Sampson, Robert J., and Corina Graif. 2009. "Neighborhood Social Capital as Differential Social Organization: Resident and Leadership Dimensions." *American Behavioral Scientist 52* (11):1579–605. doi: 10.1177/0002764209331527.
Sharkey, Patrick. 2013. *Stuck in Place: Urban Neighborhoods and the End of Progress Toward Racial Equality*. Chicago: University of Chicago Press.
Silver, Daniel Aaron, and Terry Nichols Clark. 2016. *Scenescapes: How Qualities of Place Shape Social Life*. Chicago: University of Chicago Press.
Small, Mario L., and Laura Adler. 2019. "The Role of Space in the Formation of Social Ties." *Annual Review of Sociology 45* (1):111–32. doi: 10.1146/annurev-soc-073018-022707.
Small, Mario L., Armin Akhavan, Mo Torres, and Qi Wang. 2021. "Banks, Alternative Institutions and the Spatial–Temporal Ecology of Racial Inequality in US Cities." *Nature Human Behaviour 5* (12):1622–8. doi: 10.1038/s41562-021-01153-1.
Small, Mario Luis. 2006. "Neighborhood Institutions as Resource Brokers: Childcare Centers, Interorganizational Ties, and Resource Access among the Poor." *Social Problems 53* (2):274–92. doi: 10.1525/sp.2006.53.2.274.
Small, Mario Luis. 2010. *Unanticipated Gains: Origins of Network Inequality in Everyday Life*. New York: Oxford University Press.
Stern, Mark J., Susan, C. Seifert, and Domenic Vitiello. 2010. "Migrants and the Transformation of Philadelphia's Cultural Economy." Pp. 23–51 in *Arts in the Lives of Immigrant Communities in the United States*, edited by P. DiMaggio and P. Fernandes-Kelly. New Brunswick, NJ: Rutgers University Press.
Tadmon, Daniel, and Peter S. Bearman. 2023. "Differential Spatial-Social Accessibility to Mental Health Care and Suicide." *Proceedings of the National Academy of Sciences of the United States of America 120* (19):e2301304120. doi: 10.1073/pnas.2301304120.
Tao, Zhuolin, Yang Cheng, Qingjing Zheng, and Guicai Li. 2018. "Measuring Spatial Accessibility to Healthcare Services with Constraint of Administrative Boundary: A Case Study of Yanqing District, Beijing, China." *International Journal for Equity in Health 17* (1):7. doi: 10.1186/s12939-018-0720-5.
U.S. Census Bureau. 2024. "Geography Program Glossary." https://www.census.gov/programs-surveys/geography/about/glossary.html
Vertovec, Steven. 2007. "Super-Diversity and Its Implications." *Ethnic and Racial Studies 30* (6):1024–54. doi: 10.1080/01419870701599465.
Wilson, William Julius. 1987. *The Truly Disadvantaged: The Inner City, the Underclass, and Public Policy*. Chicago: University of Chicago Press.
Winkler, Richelle, L. Jeffrey, A. G. Clark, Dexter H. Locke, Peleg Kremer, Myla F. J. Aronson, Fushcia-Ann Hoover, Hogyeum Evan Joo, Daniele La Rosa, KangJae Jerry Lee, Susannah B., et al. 2024. "Unequal Access to Social, Environmental and Health Amenities in US Urban Parks." *Nature Cities 1* (12):861–70.,. doi: 10.1038/s44284-024-00153-2.
Zapatka, Kasey, and Van C. Tran. 2023. "New Frontiers of Integration: Convergent Pathways of Neighborhood Diversification in Metropolitan New York." *RSF: The Russell Sage Foundation Journal of the Social Sciences 9* (1):52–83. doi: 10.7758/RSF.2023.9.1.03.
Zhang, Lihong, Xiang-Yu Hou, and Yan Liu. 2025. "Measuring Mental Health Service Accessibility for Indigenous Populations: A Systematic Review." *Journal of Racial and Ethnic Health Disparities 12* (1):594–612. doi: 10.1007/s40615-023-01899-6.

Zilberstein, Shira. 2019. "Space Making as Artistic Practice: The Relationship between Grassroots Art Organizations and the Political Economy of Urban Development." *City & Community 18* (4):1142–61. doi: 10.1111/cico.12458.
Zukin, Sharon. 1982. *Loft Living: Culture and Capital in Urban Change*. New Brunswick, NJ: Rutgers University Press.
Zukin, Sharon. 1996. *The Cultures of Cities*. Cambridge, MA: Wiley-Blackwell.
Zukin, Sharon, Valerie Trujillo, Peter Frase, Danielle Jackson, Tim Recuber, and Abraham Walker. 2009. "New Retail Capital and Neighborhood Change: Boutiques and Gentrification in New York City." *City & Community 8* (1):47–64. doi: 10.1111/j.1540-6040.2009.01269.x.

Appendix

1. Classification of ethnic-oriented and citywide organizations

We leveraged the GPT 4-o model to classify organizations using a rule-based, multiclass keyword scoring algorithm with two labels: *ethnic* and *citywide*. For each organization, the input text combined the organization name, discipline, and scraped mission statement. All texts are lowercased before score matching. We then applied predefined keyword dictionaries for each label, and computed scores as keyword-match counts. Organizations were assigned to the label with the highest score; if both scores are zero, they were labeled as unclassified with respect to these two types. These unmatched organizations were retained in the full organization sample, which we treat as community-based by default.

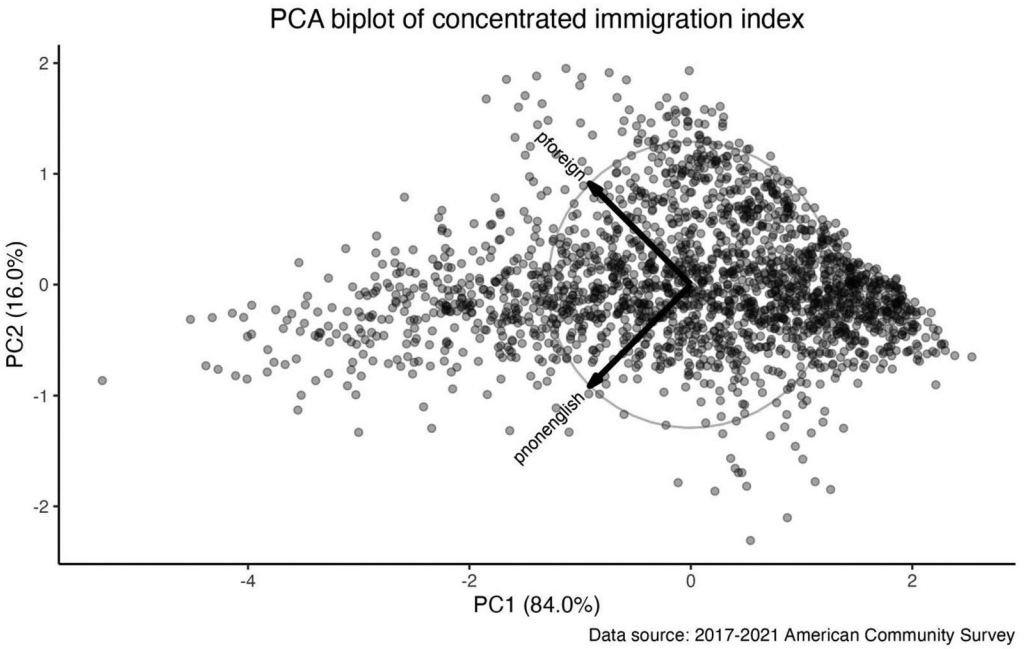
Keyword dictionary for *ethnic* label:

"ethnic": [
"african", "latino", "latina", "hispanic", "asian", "ethnic",
"diaspora", "cultural heritage", "multicultural", "immigrant",
"chinese", "korean", "japanese", "vietnamese", "filipino", "indian",
"pakistani", "bangladeshi", "thai", "cambodian", "laotian",
"puerto rican", "dominican", "mexican", "chicano", "chicana",
"salvadoran", "guatemalan", "colombian", "ecuadorian", "peruvian",
"cuban", "honduran", "nicaraguan", "venezuelan",
"caribbean", "haitian", "jamaican", "trinidadian", "guyanese",
"irish", "italian", "polish", "greek", "russian", "ukrainian",
"middle eastern", "arab", "persian", "armenian", "turkish",
"french", "german", "spanish", "portuguese", "swedish", "norwegian",
"danish", "finnish", "dutch", "belgian", "austrian", "swiss",
"hungarian", "czech", "slovak", "croatian", "serbian", "bosnian",
"slovenian", "bulgarian", "romanian", "albanian", "macedonian",
"latvian", "lithuanian", "estonian",
"black", "african american", "afro", "schomburg",
"harlem renaissance", "black arts", "indigenous", "native", "tribal",
"muslim", "jewish", "polynesian", "pacific islander"
]

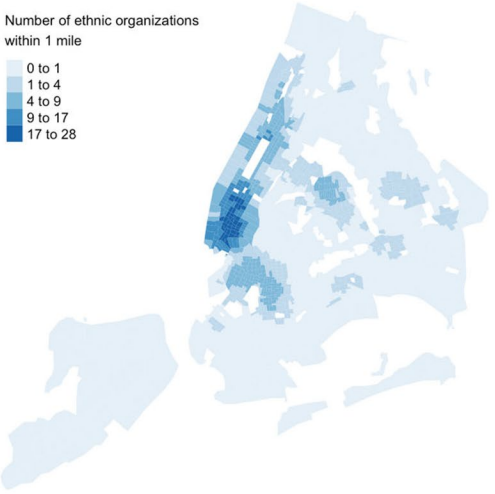
Keyword dictionary for *citywide* label:

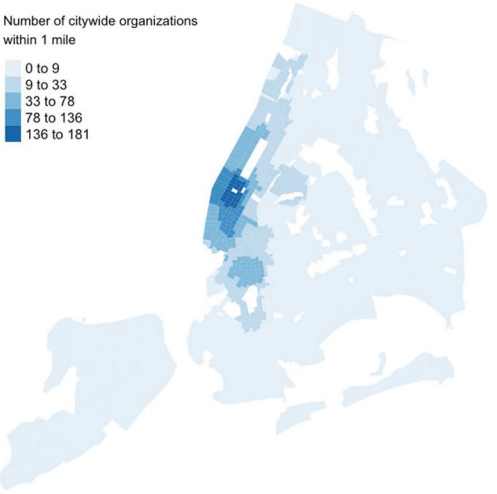
"citywide": [
"metropolitan", "met", "institute", "society",
"academy", "conservatory", "philharmonic", "symphony", "opera",
"ballet", "theater", "theatre", "museum", "fine arts",
"classical", "orchestra", "chamber music",
"lincoln center", "carnegie", "juilliard", "brooklyn academy",
"moma", "whitney", "guggenheim", "frick", "morgan library",
"library", "botanic", "botanical", "education",
"learning", "student", "public", "parks", "recreation",
"city", "municipal", "regional", "world-class", "international", "major"
]

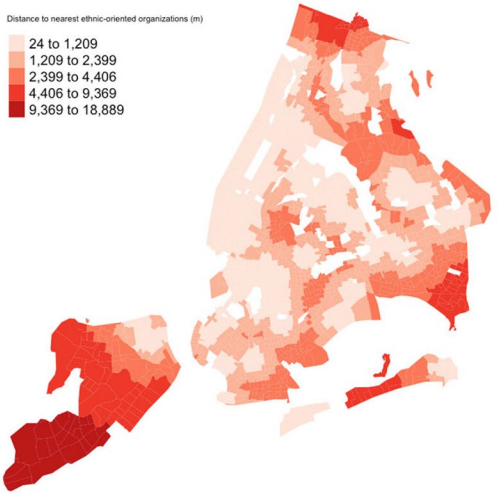
2. Principal component analysis (PCA) biplot for concentrated immigration index

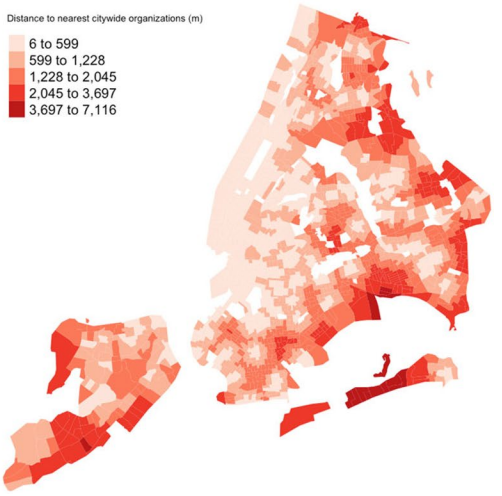


3. Additional maps for ethnic-oriented and citywide organizations



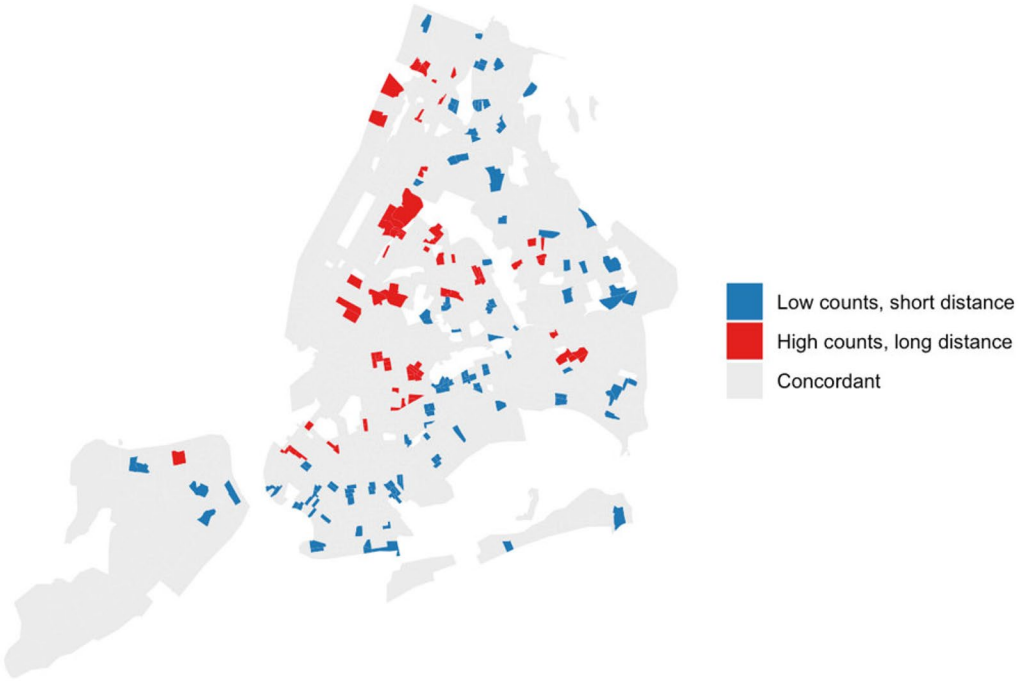






4. Sensitivity analysis of discordance between two cultural accessibility measures

To assess whether Figures 2 and 3 represent different spatial patterns, we constructed a tract-level discordance map using the two cultural accessibility measures. Because shorter nearest distance indicates higher access, we first reverse-coded distance so that larger values indicate greater proximity. We then converted both measures into quantiles and mapped their joint distribution in bivariate color scheme. The map identifies where the two measures diverge: ***Low counts, short distance*** are tracts in the bottom two quintiles of count-based access but top two quintiles of proximity-based access; ***high counts, long distance*** are tracts in the top two quintiles of count-based access but bottom two quintiles of proximity-based access; ***concordant*** are tracts where both indicators reflect similarly high or similarly low access. This analysis shows that the two measures have overlapping broad patterns but are not interchangeable. They capture distinct dimensions of accessibility: low availability versus minimum proximity.



5. Sensitivity analysis for regression models with spatial lags

For robustness, we estimated spatial lag (SAR) models using a queen-contiguity weights matrix for two outcomes: number of organizations within 1 mile and average distance to the nearest organization. As both dependent variables (buffer-based counts and nearest-neighbor distance) are already spatially constructed, they inherently capture local spillover processes. Consistent with this, the SAR lag parameters for both outcomes are significant. The estimated effects remain directionally consistent with OLS models but are less precise after including the spatial lag term. This suggests that the lag term absorbs broad spatial clustering already embedded in the outcomes. We therefore report SAR estimates as a sensitivity check for residual spatial dependence and retain OLS as the primary models for interpretability.

	Number of organizations within 1 mile		Distance to nearest organization (m)	
	Model 1	Model 2	Model 3	Model 4
Predominantly Black	1.1	0.7	−18.1	−32.1
	(0.6)	(0.7)	(17.2)	(18.8)
Predominantly Hispanic	0.6	0.3	−10.2	−25.0
	(0.7)	(0.8)	(19.5)	(21.6)
Predominantly Asian	1.3	1.2	2.6	−3.4
	(1.1)	(1.1)	(30.2)	(30.0)
White mixed	0.5	0.6	−23.8	−21.9
	(0.6)	(0.6)	(17.2)	(17.1)
Non-White mixed	1.4	1.2	6.2	−0.4
	(0.7)	(0.7)	(19.4)	(19.7)
Concentrated immigration	0.2	0.1	−7.8	−9.7
	(0.2)	(0.2)	(4.9)	(5.2)
% Residents in poverty		−0.03		0.4
		(0.02)		(0.4)
% Residents with college degree		−0.04*		−0.3
		(0.02)		(0.5)
County-level mobility		0.1		−2.3
		(0.1)		(2.8)
Lag: organizations within 1 mile	1.0***	1.0***		
	(0.003)	(0.004)		
Lag: distance to nearest organization			1.1***	1.1***
			(0.01)	(0.01)
Constant	−1.5**	0.9	−21.0	−7.2
	(0.5)	(1.2)	(16.3)	(38.5)
R-squared	0.981	0.981	0.773	0.777
Adjusted R-squared	0.981	0.981	0.772	0.776
N	2,107	2,105	2,107	2,105

*$p < .05$; **$p < .01$; ***$p < .001$.